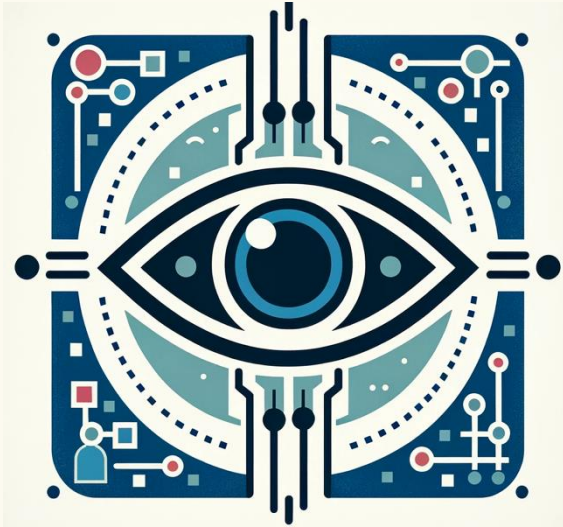




COMPUTER VISION

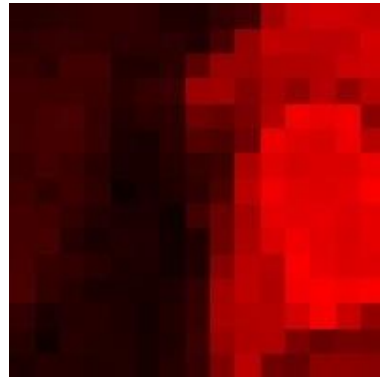
Filtering

What is an image?

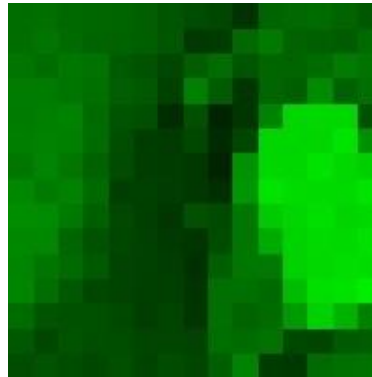


color image patch

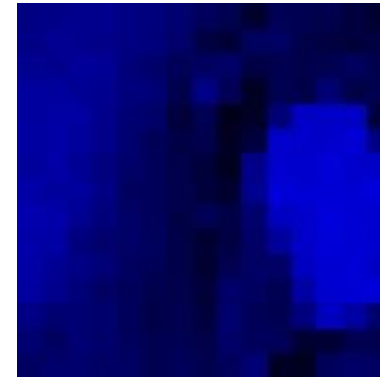
red



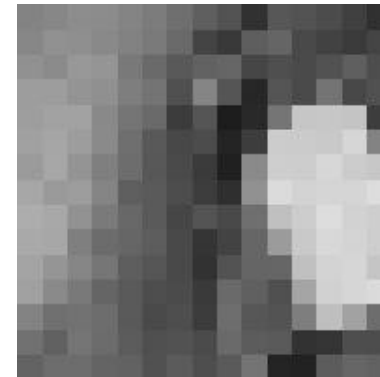
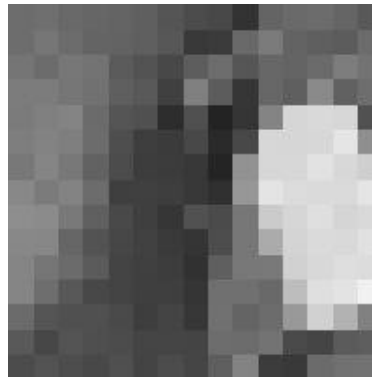
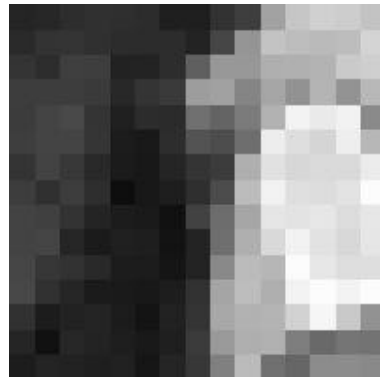
green



blue



colorized for visualization



actual intensity values per channel

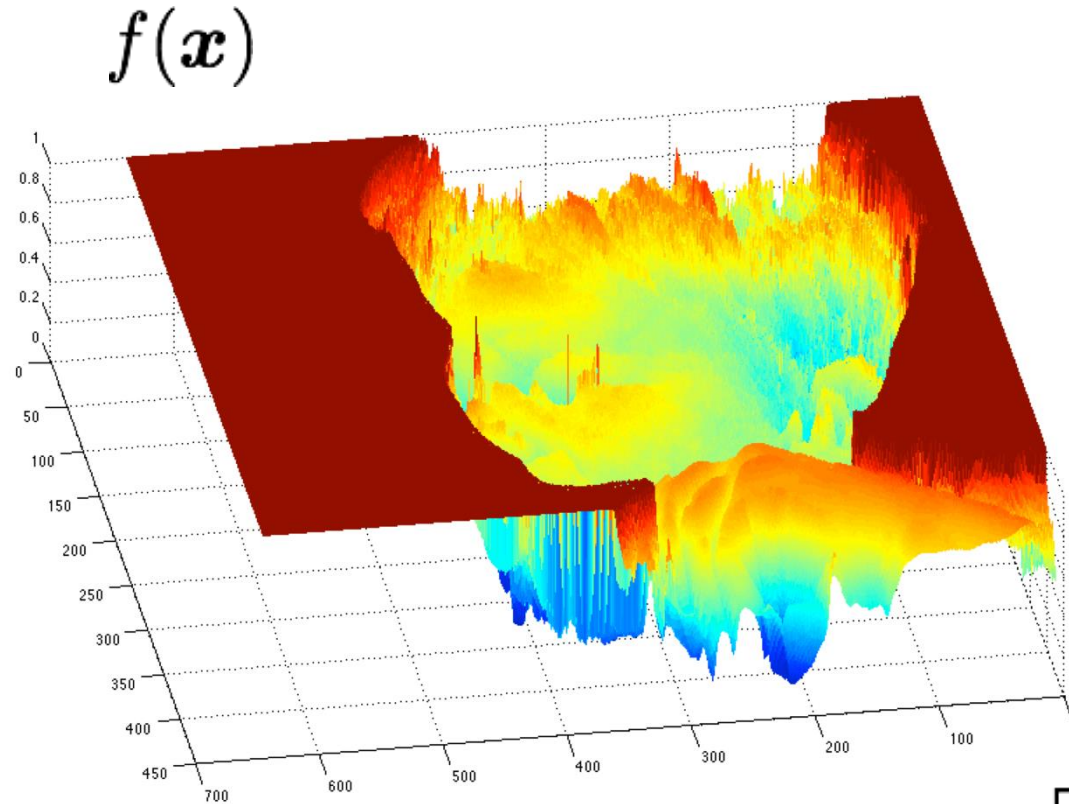
Each channel
is a 2D array of
numbers

What is an image?



grayscale image

What is the range of the image function f ?



A (grayscale) image is a 2D function

domain $\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$

What types of image transformations can we do?



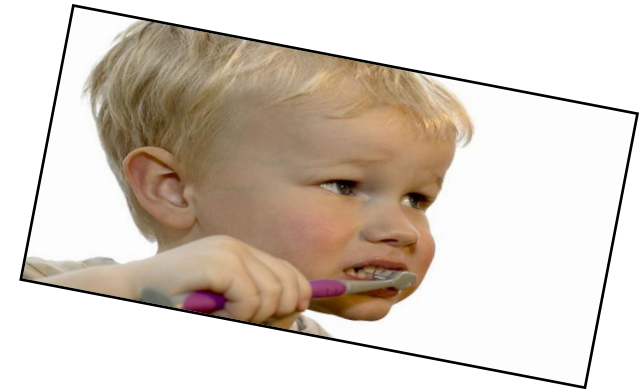
Filtering



changes pixel *values*



Warping



changes pixel *locations*

What types of image transformations can we do?

F



Filtering



$$G(\mathbf{x}) = h\{F(\mathbf{x})\}$$

G



changes pixel *values*

F

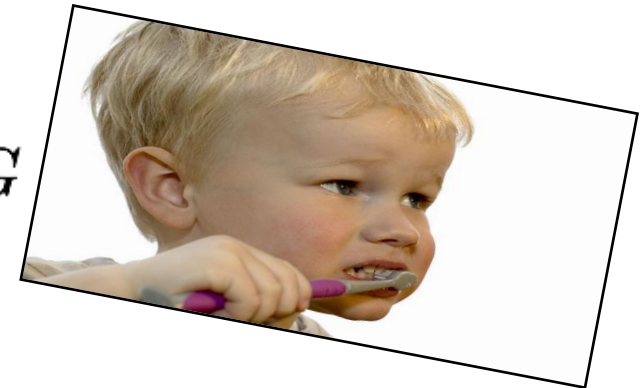


Warping



$$G(\mathbf{x}) = F(h\{\mathbf{x}\})$$

G

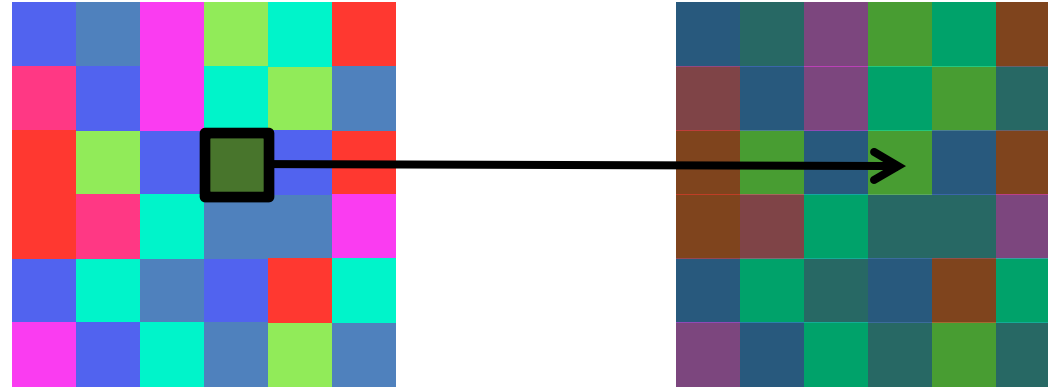


changes pixel *locations*



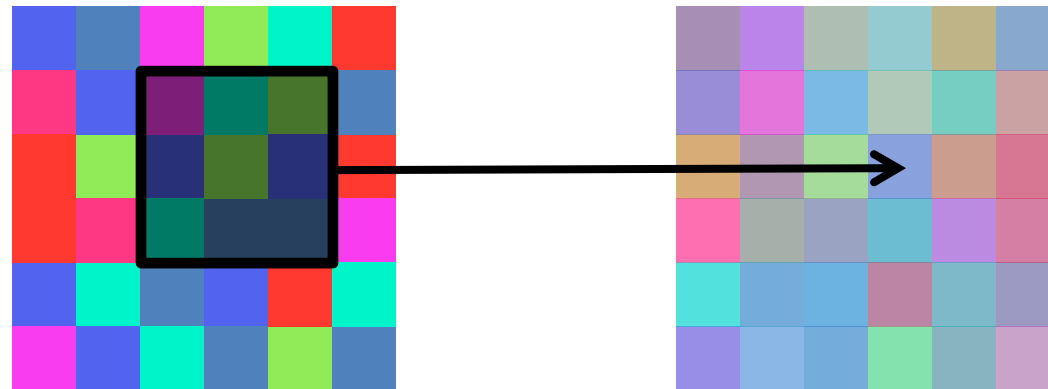
What types of image filtering can we do?

Point Operation



point processing

Neighborhood Operation



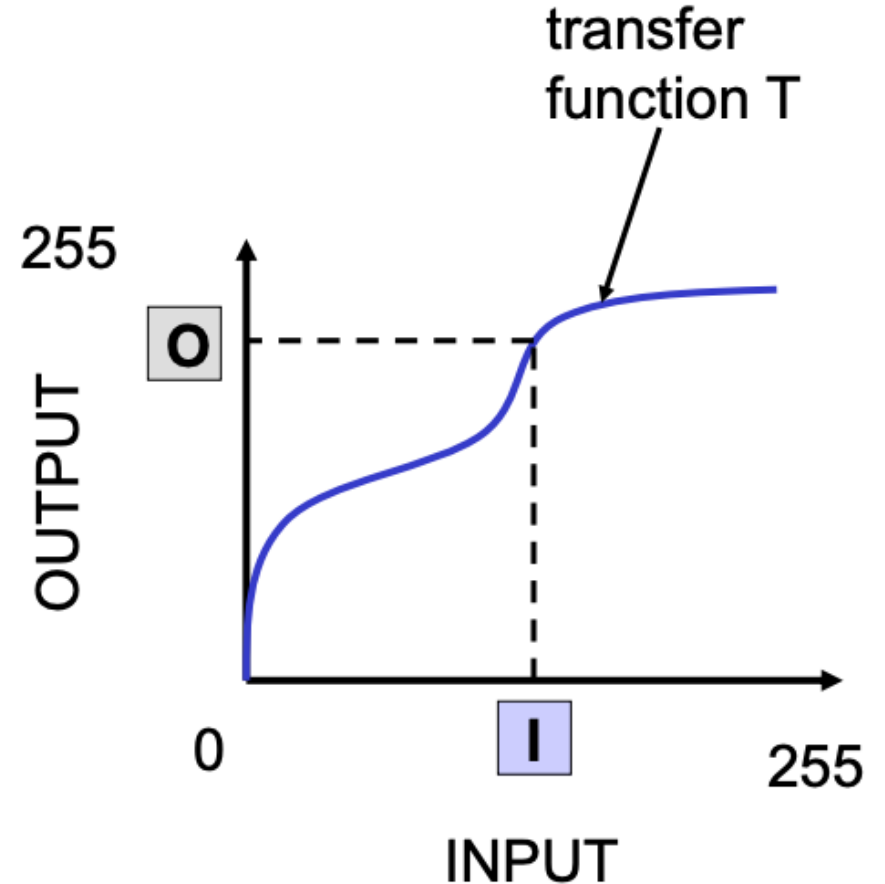
“filtering”

Point processing

Point processing – general idea

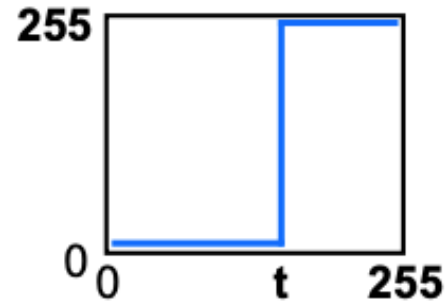
$$O = T(I)$$

- Input pixel value, I , mapped to output pixel value, O , via **transfer function T**
- T is a **point-to-point transformation**, i.e. only information at $I(x,y)$ used to generate $O(x,y)$
- Typically applied in intensity, but can be applied to the colour channels → it may distort the colours (RGB)



Thresholding

$$O(x,y) = \begin{cases} I_{\max} & \text{if } I(x,y) > t \\ I_{\min} & \text{if } I(x,y) \leq t \end{cases}$$



How can we define t ?

→ Fixed value or adaptive



Color image



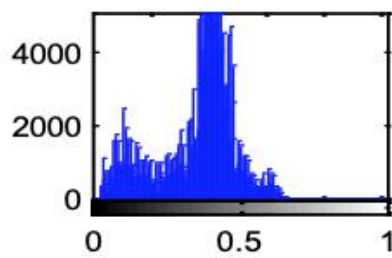
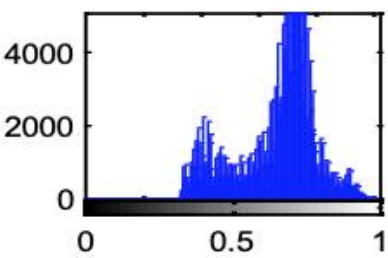
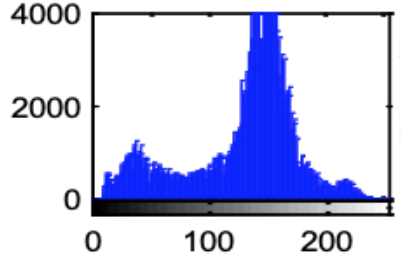
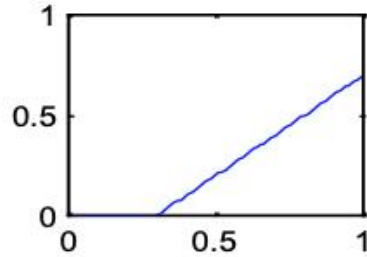
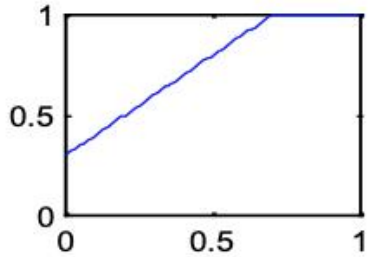
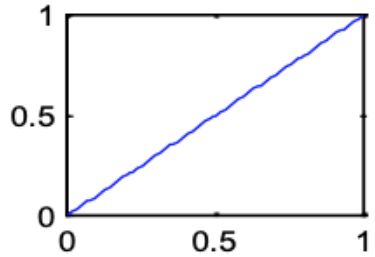
Graylevel image



Thresholded graylevel image ($t=89$)

Linear transformations

Brightness



Linear stretch

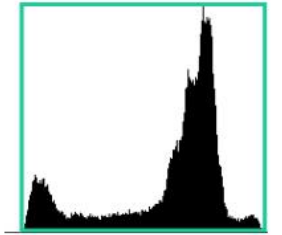
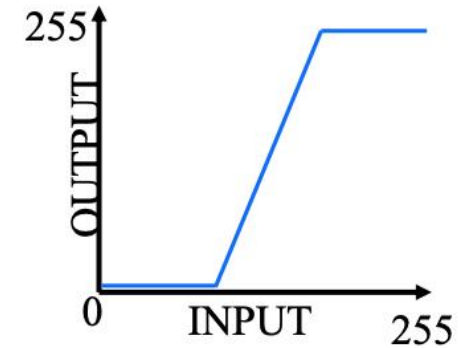
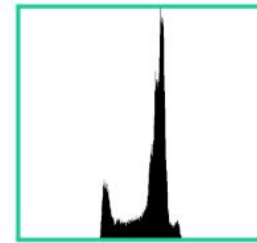
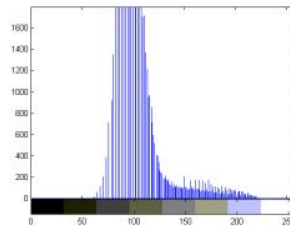
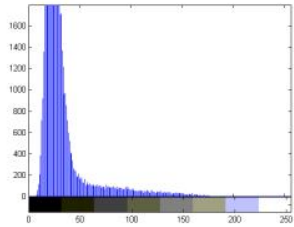
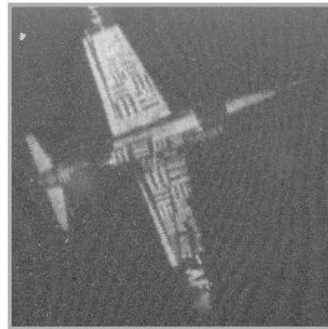
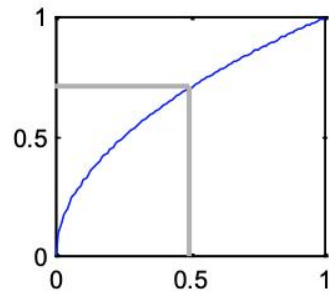
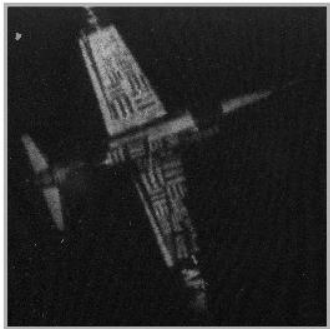


Image histograms

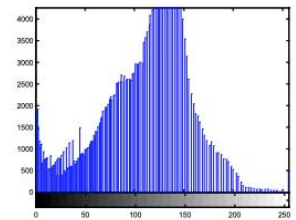
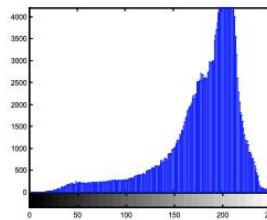
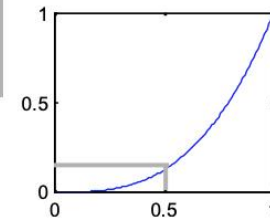
Non-linear scaling

$$O = I^\gamma$$



$$\gamma = 0.5$$

enhances contrast in dark regions



$$\gamma = 3$$

enhances contrast in bright regions

Examples of point processing

original



x

darken



lower contrast



non-linear lower contrast



invert



lighten



raise contrast



non-linear raise contrast



How would you implement these?

Examples of point processing

original



$$x$$

darken



$$x - 128$$

lower contrast



non-linear lower contrast



invert



lighten



raise contrast



non-linear raise contrast



Examples of point processing

original



$$x$$

darken



$$x - 128$$

lower contrast



$$\frac{x}{2}$$

non-linear lower contrast



invert



lighten



raise contrast



non-linear raise contrast



Examples of point processing

original



$$x$$

darken



$$x - 128$$

lower contrast



$$\frac{x}{2}$$

non-linear lower contrast



$$\left(\frac{x}{255}\right)^{1/3} \times 255$$

invert



lighten



raise contrast



non-linear raise contrast



Examples of point processing

original



$$x$$

darken



$$x - 128$$

lower contrast



$$\frac{x}{2}$$

non-linear lower contrast



$$\left(\frac{x}{255}\right)^{1/3} \times 255$$

invert



$$255 - x$$

lighten



raise contrast



non-linear raise contrast



Examples of point processing

original



$$x$$

darken



$$x - 128$$

lower contrast



$$\frac{x}{2}$$

non-linear lower contrast



$$\left(\frac{x}{255}\right)^{1/3} \times 255$$

invert



$$255 - x$$

lighten



$$x + 128$$

raise contrast



non-linear raise contrast



Examples of point processing

original



$$x$$

darken



$$x - 128$$

lower contrast



$$\frac{x}{2}$$

non-linear lower contrast



$$\left(\frac{x}{255}\right)^{1/3} \times 255$$

invert



$$255 - x$$

lighten



$$x + 128$$

raise contrast



$$x \times 2$$

non-linear raise contrast



Examples of point processing

original



$$x$$

darken



$$x - 128$$

lower contrast



$$\frac{x}{2}$$

non-linear lower contrast



$$\left(\frac{x}{255}\right)^{1/3} \times 255$$

invert



$$255 - x$$

lighten



$$x + 128$$

raise contrast



$$x \times 2$$

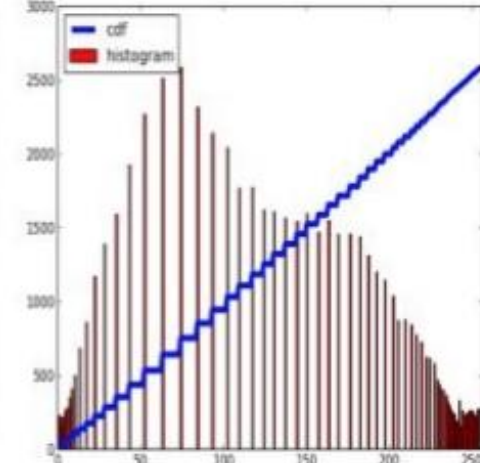
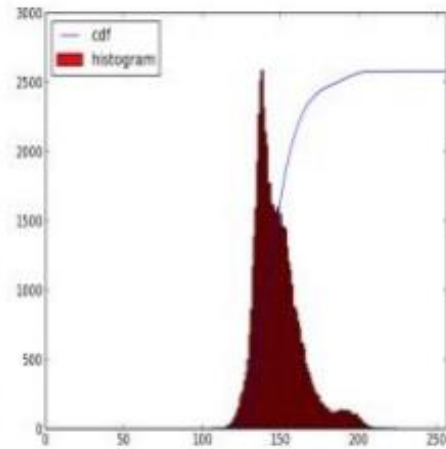
non-linear raise contrast



$$\left(\frac{x}{255}\right)^2 \times 255$$

Histogram equalization

After equalization the cumulative distribution function is almost linear



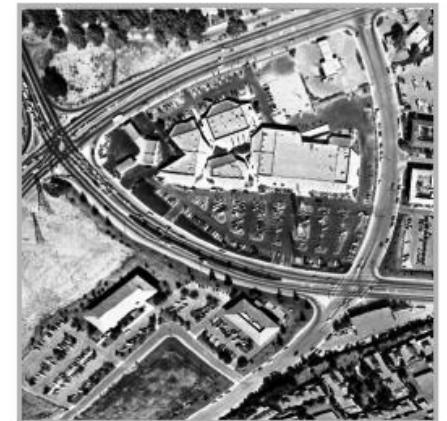
Original



$\gamma > 1$



Histogram
equalization

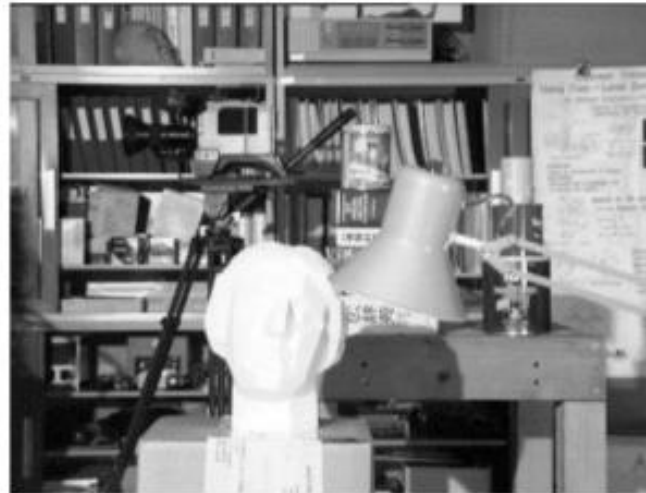


CLAHE – Contrast Limited Adaptive Histogram Equalization

- For some images, it might be preferable to apply different kinds of equalization in different regions
- Instead of computing a single curve, the image is divided into $M \times M$ pixel non-overlapped sub-blocks and **separate histogram equalization** is performed in each sub-block
- To avoid blocking artifacts (i.e., intensity discontinuities), the equalization functions are **smoothly interpolated** between blocks
- This technique is known as adaptive histogram equalization (AHE) and its contrast limited (gain-limited, to avoid noise amplification) version is known as **CLAHE**



Original image



After global equalization
(notice the effect on the face of the statue)



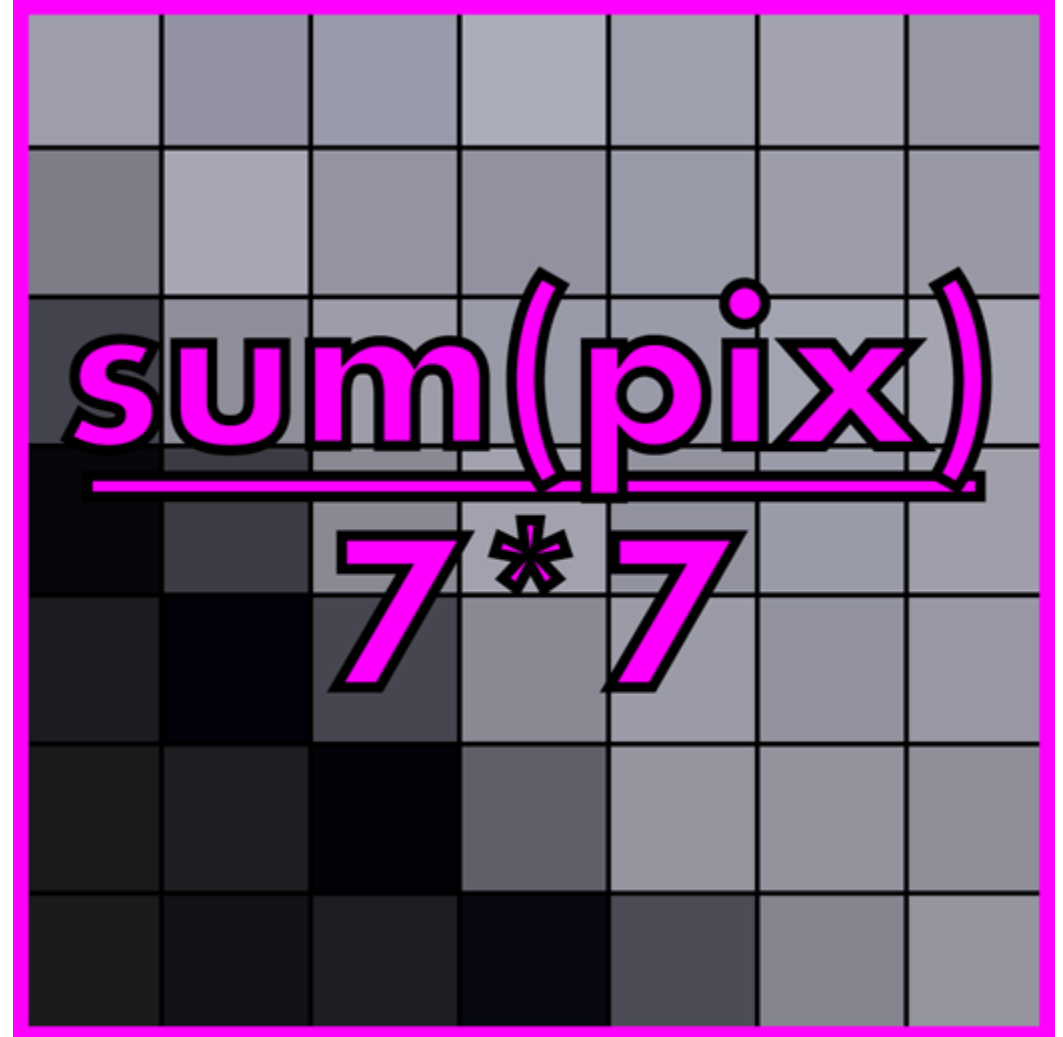
After CLAHE

Filtering

Linear shift-invariant image filtering

- Replace each pixel by a ***linear* combination of its neighbors** (and possibly itself)
- The combination is determined by the filter's ***kernel***
- The same kernel is ***shifted* to all pixel locations** so that all pixels use the same linear combination of their neighbors

Let us start with a box filter



What is averaging? A weighted sum

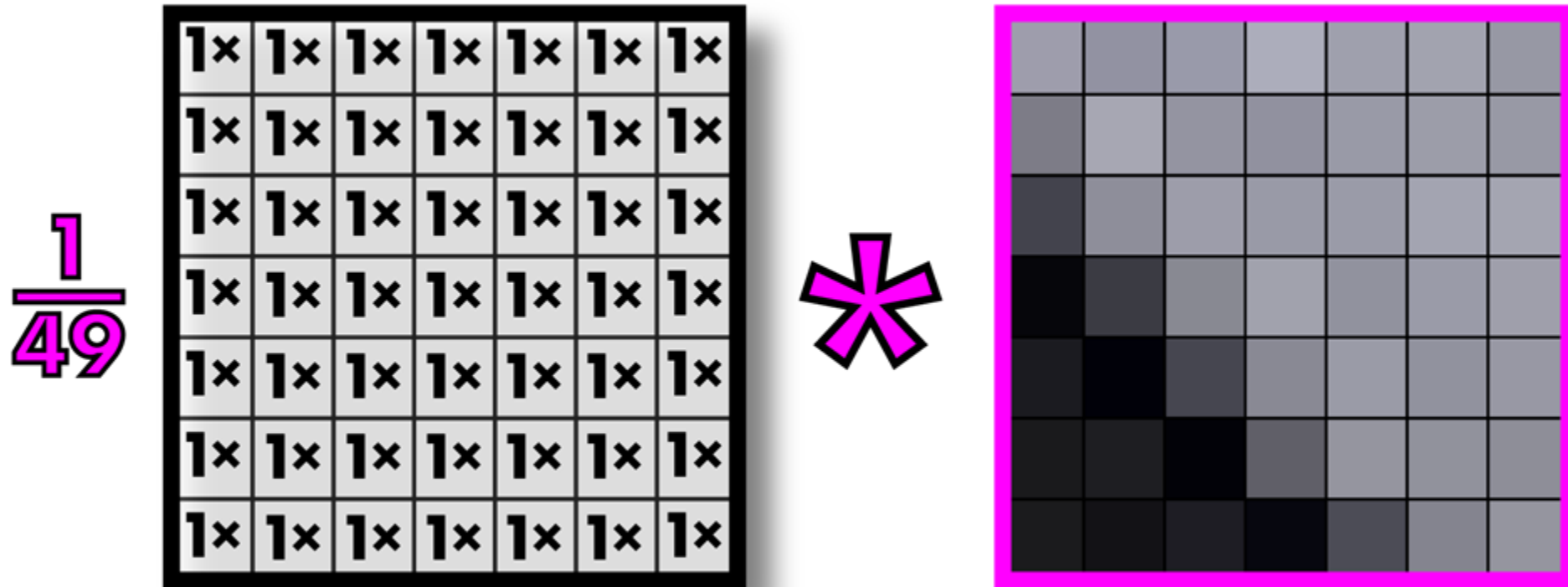
sum $\left[\frac{1}{49} \right]$

1×	1×	1×	1×	1×	1×	1×
1×	1×	1×	1×	1×	1×	1×
1×	1×	1×	1×	1×	1×	1×
1×	1×	1×	1×	1×	1×	1×
1×	1×	1×	1×	1×	1×	1×
1×	1×	1×	1×	1×	1×	1×
1×	1×	1×	1×	1×	1×	1×

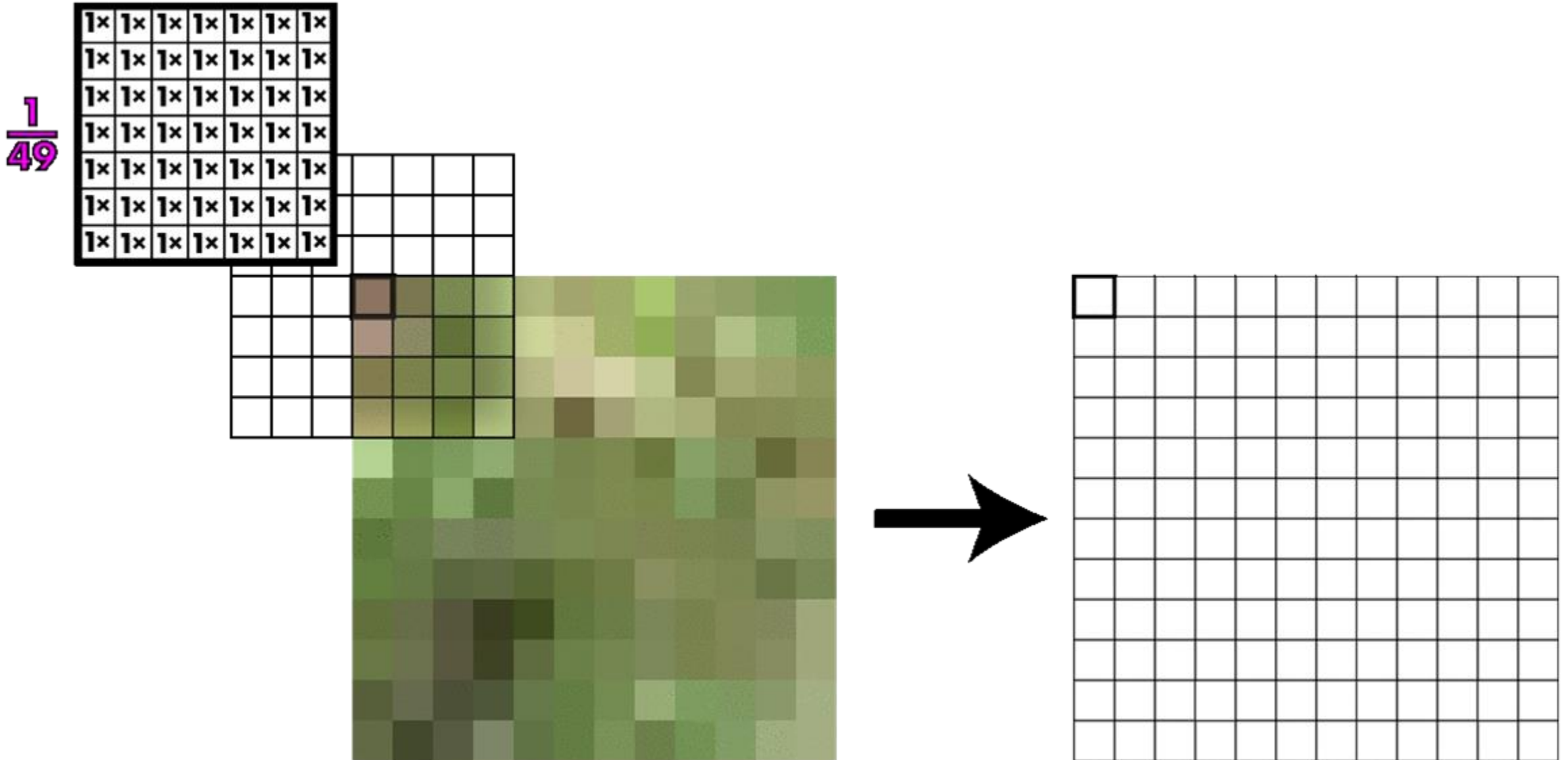
This operation is called “*convolution*”

(it actually is a cross-correlation operation)

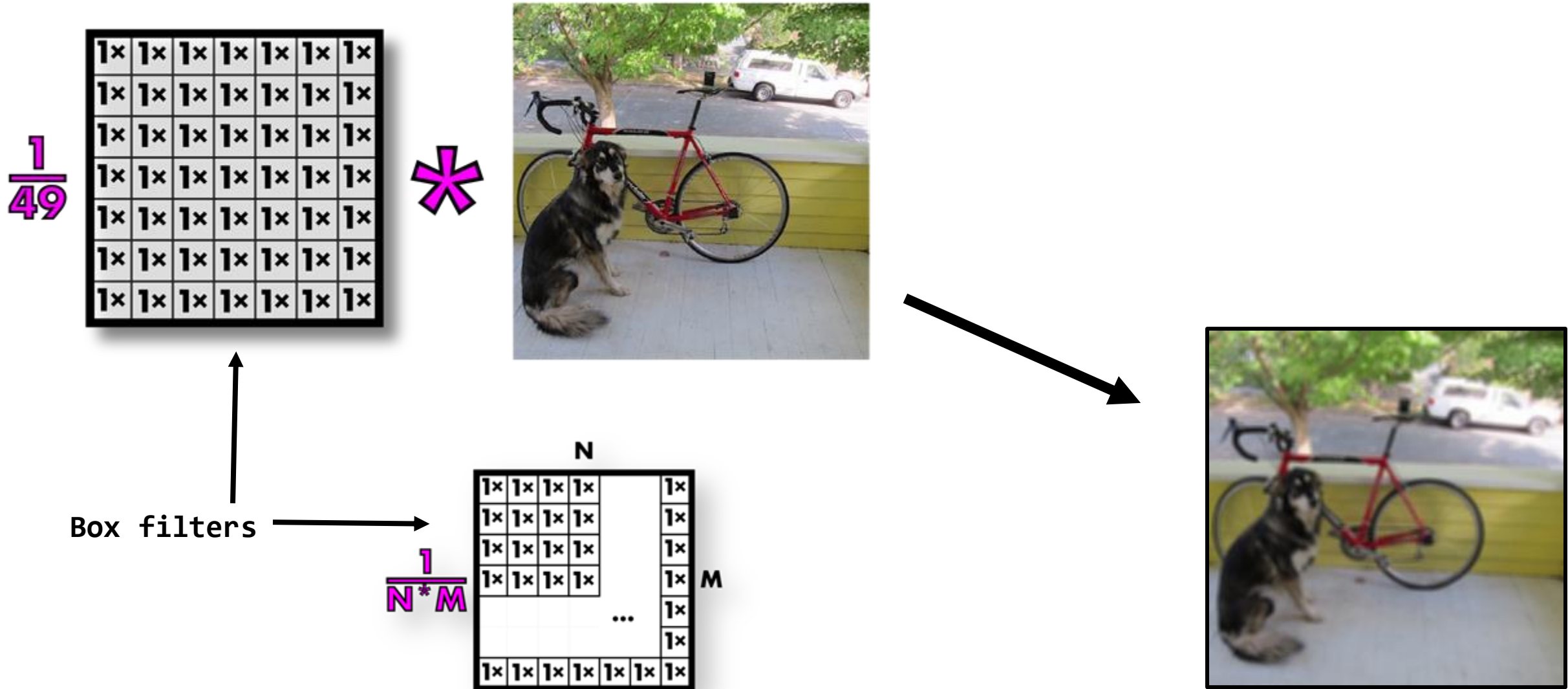
Filter or kernel



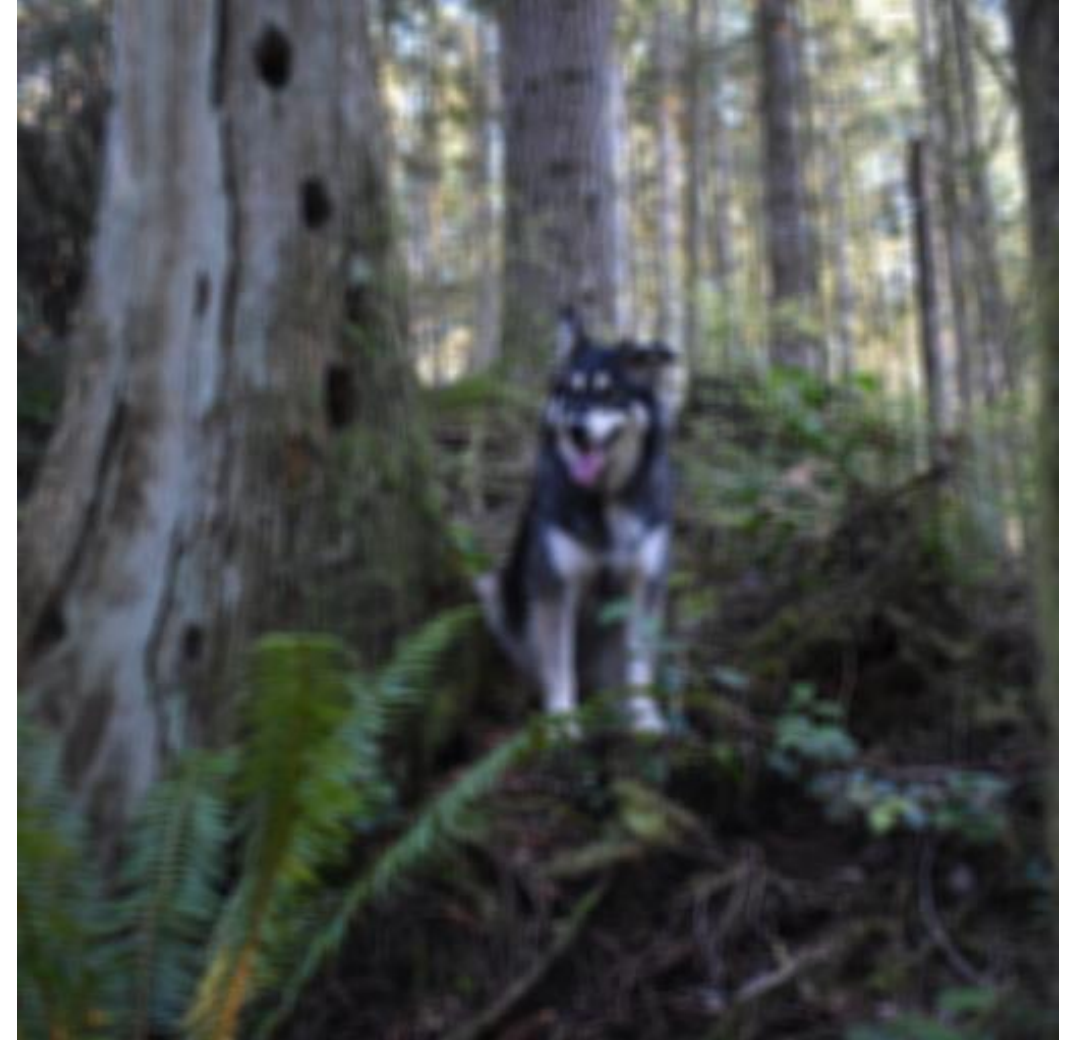
Kernel slides across image



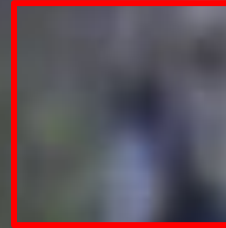
Box / Average filters smooth images



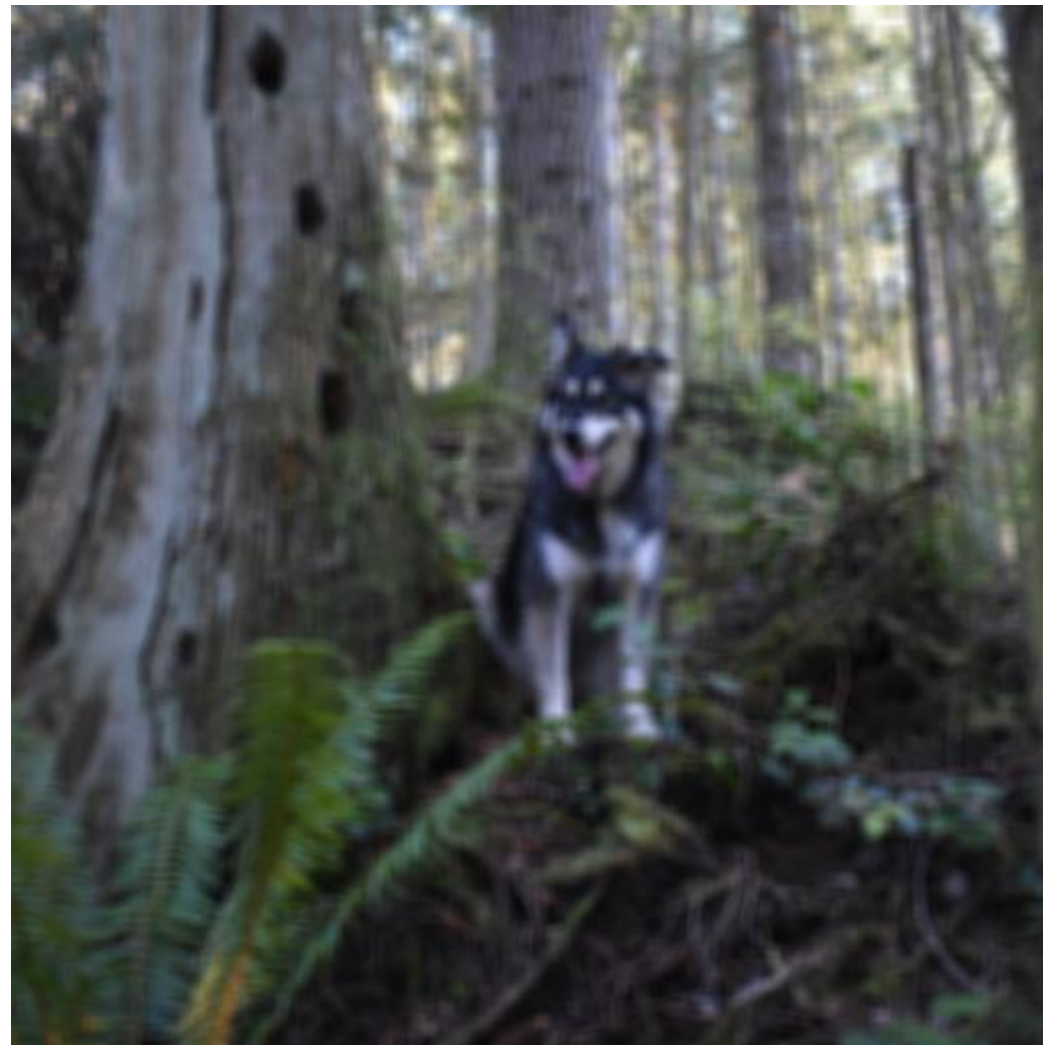
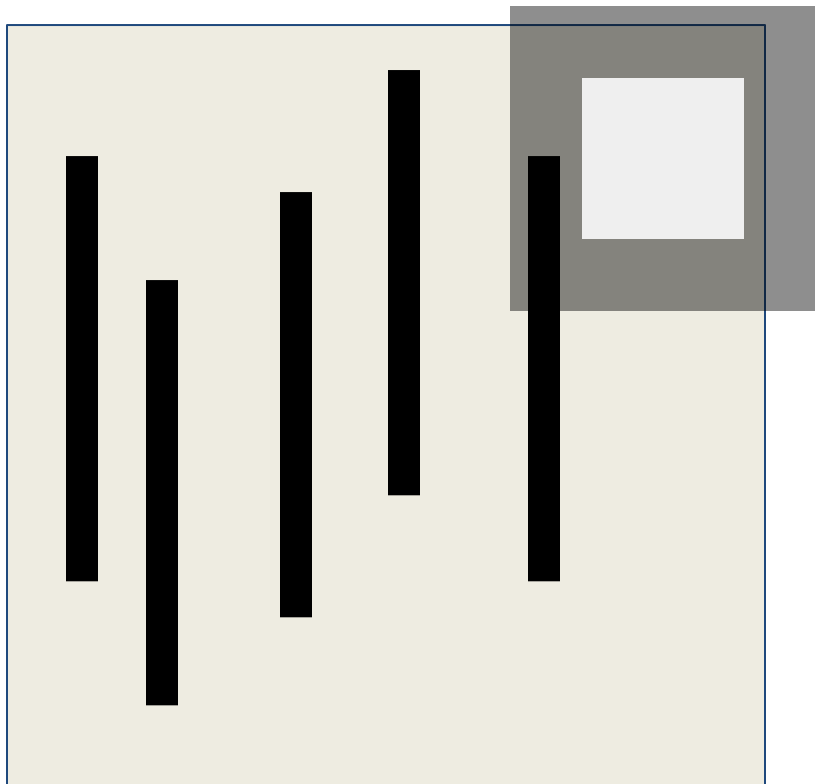
Box filters have artifacts



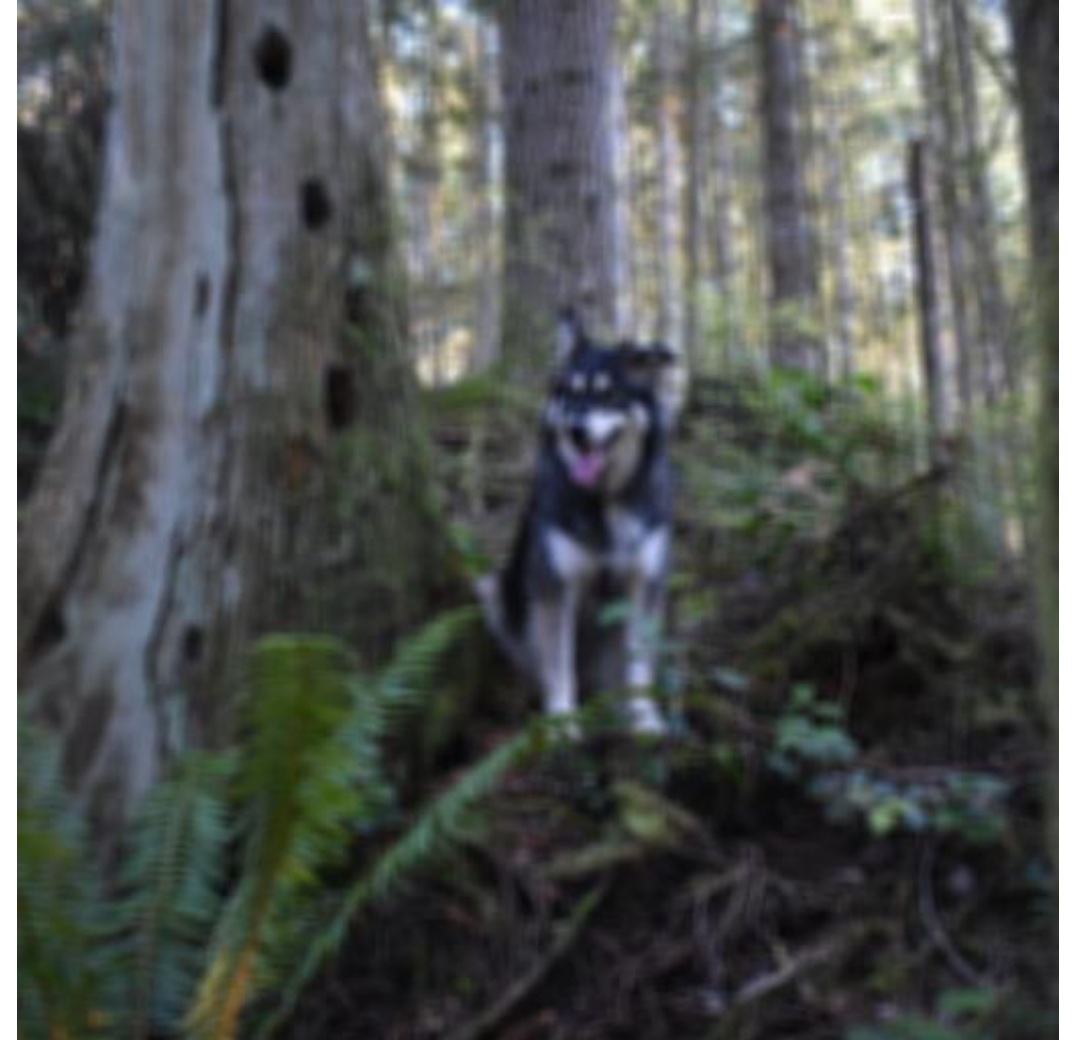
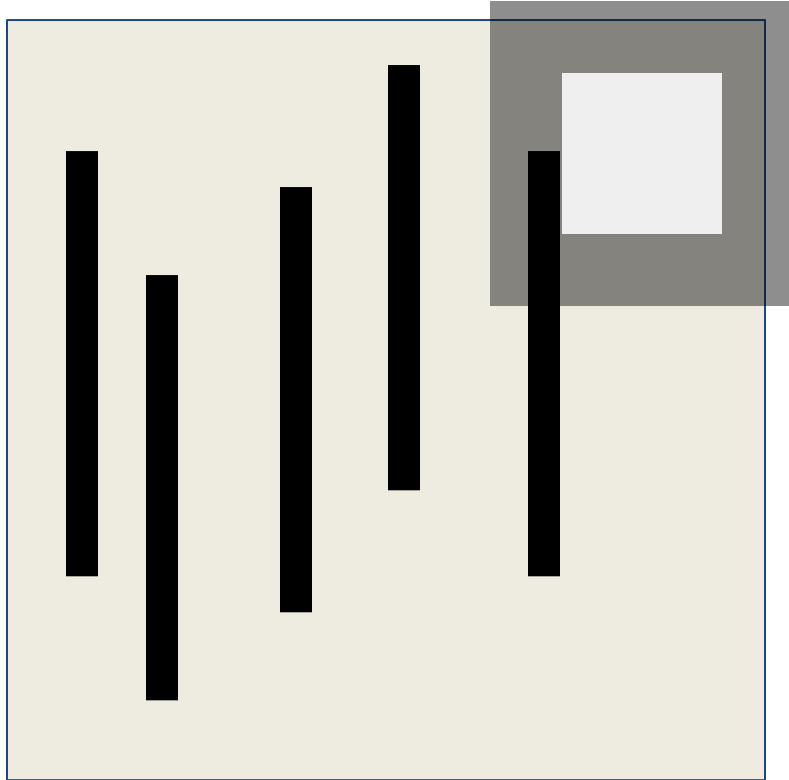
Box filters have artifacts



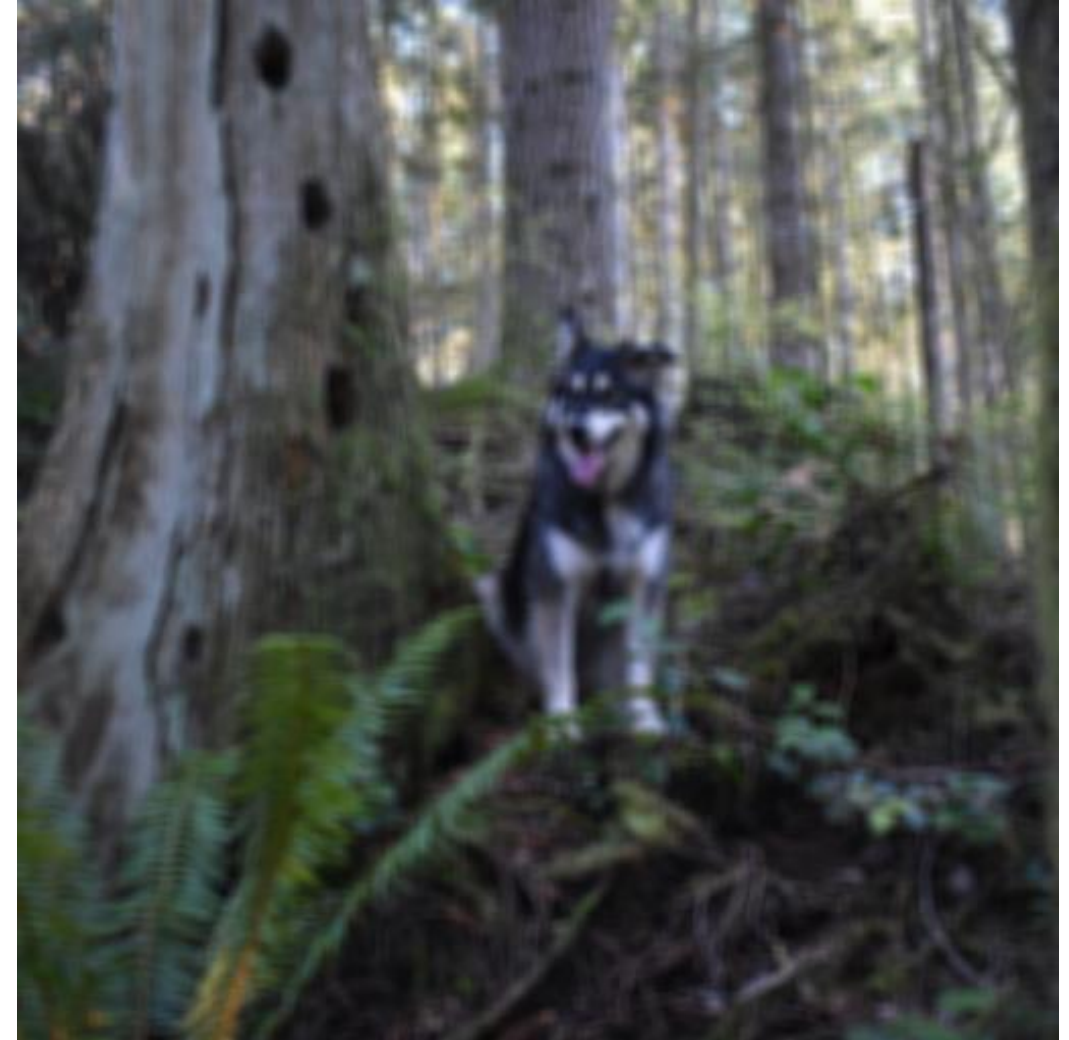
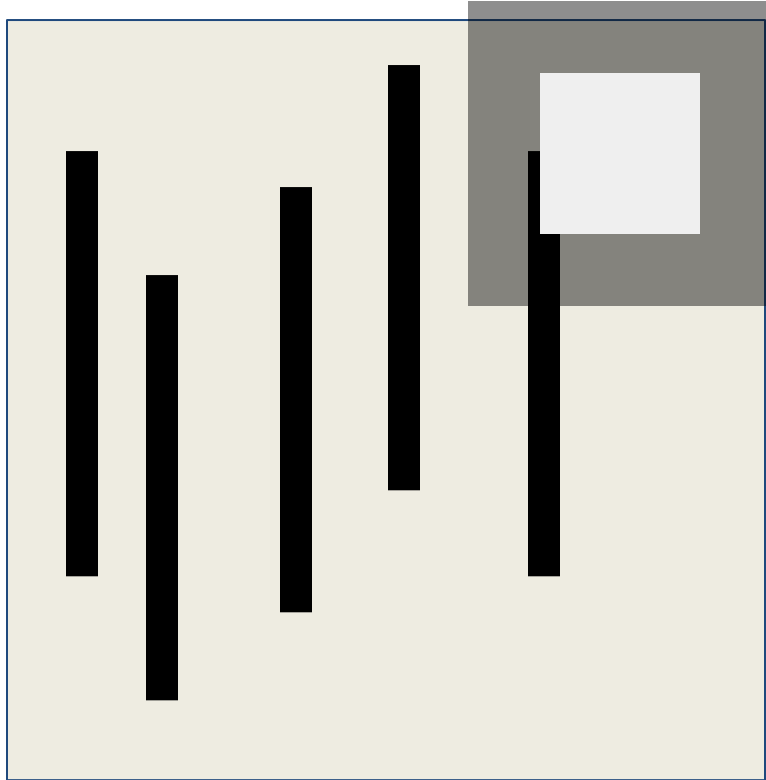
Box filters: vertical + horizontal streaking



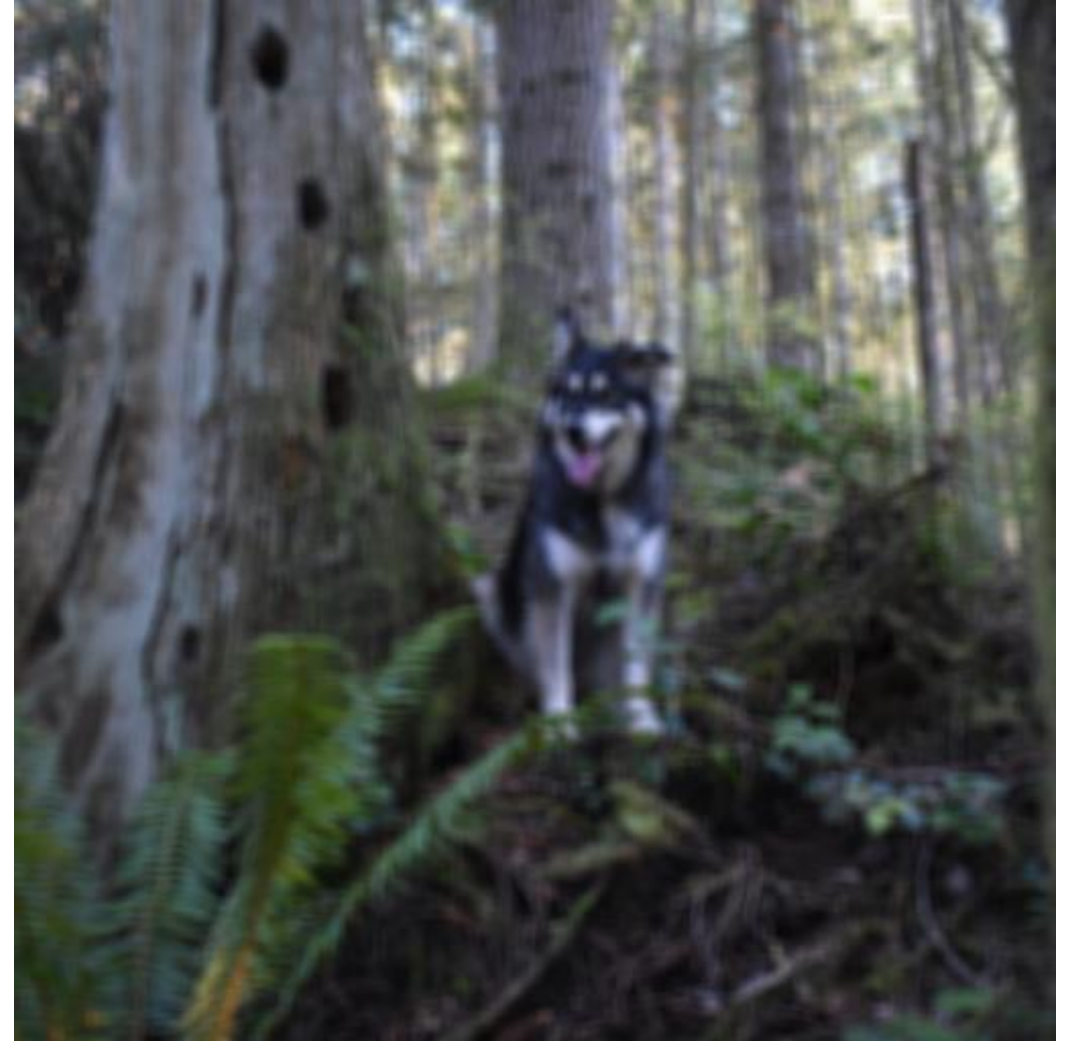
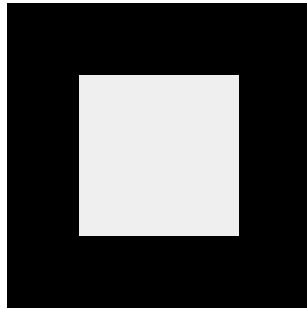
Box filters: vertical + horizontal streaking



Box filters: vertical + horizontal streaking

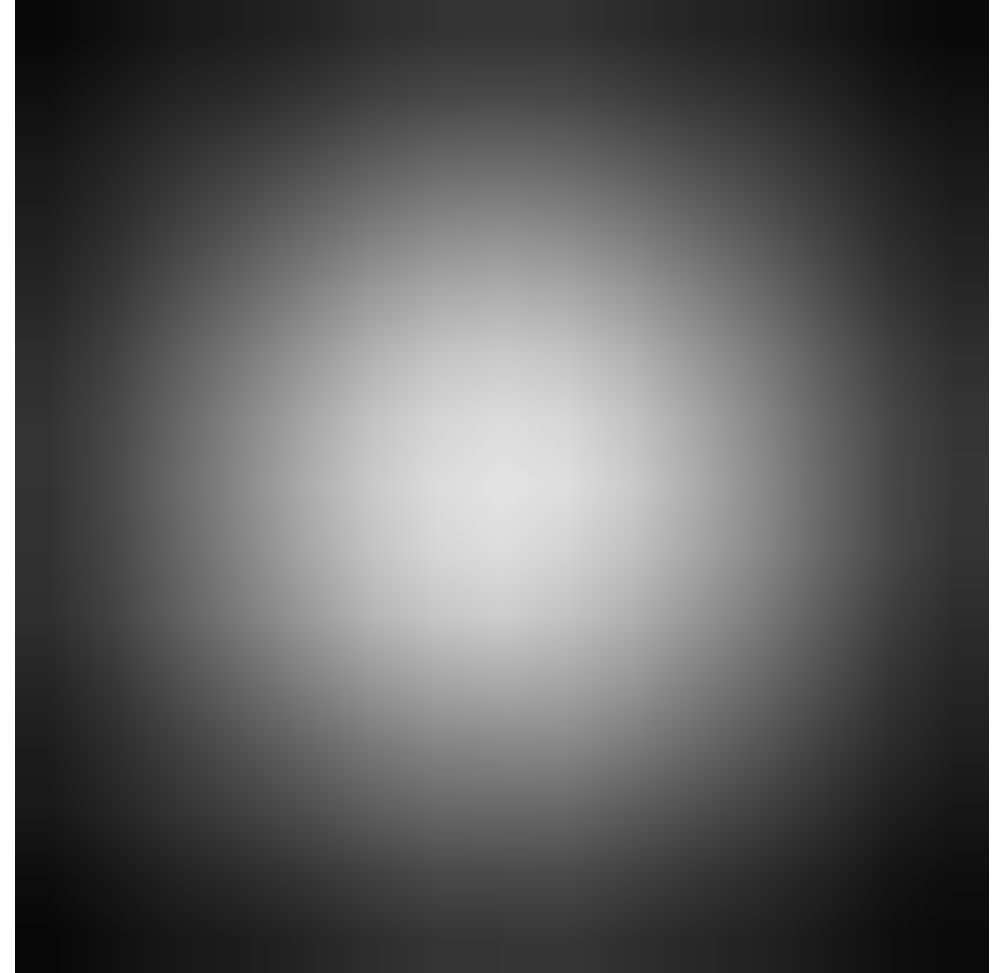
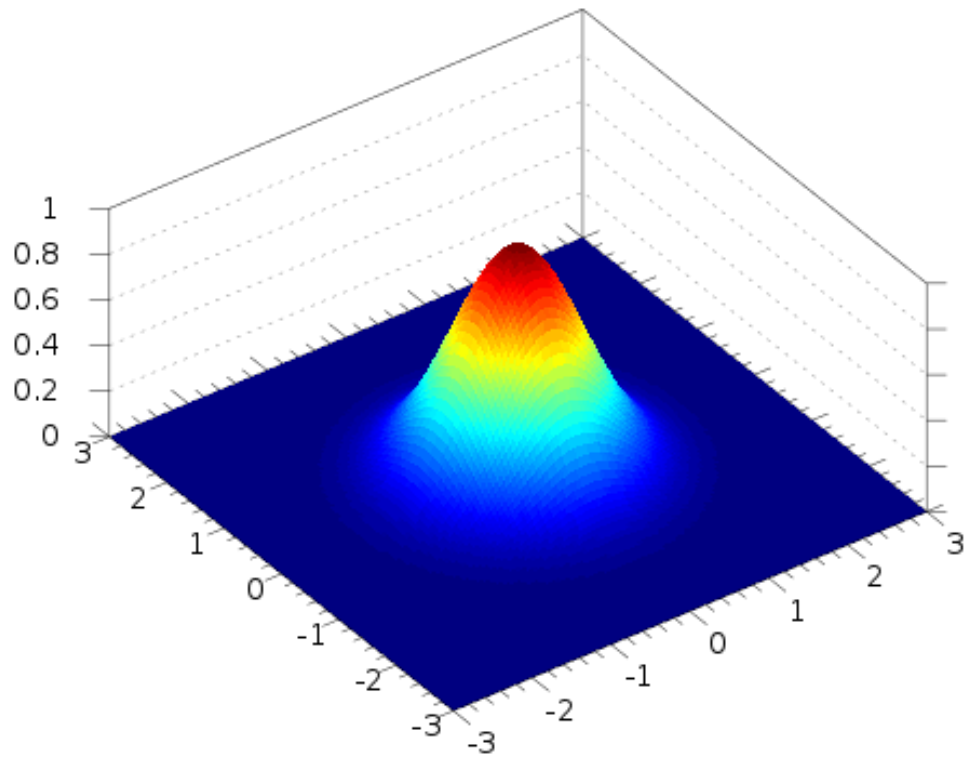


We want a smoothly weighted kernel



2D Gaussian Kernel

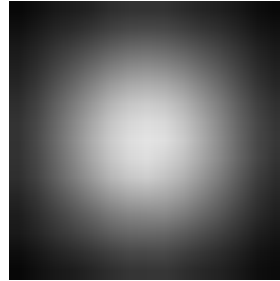
$$g(x, y) = \frac{1}{2\pi\sigma^2} \cdot e^{-\frac{x^2+y^2}{2\sigma^2}}$$



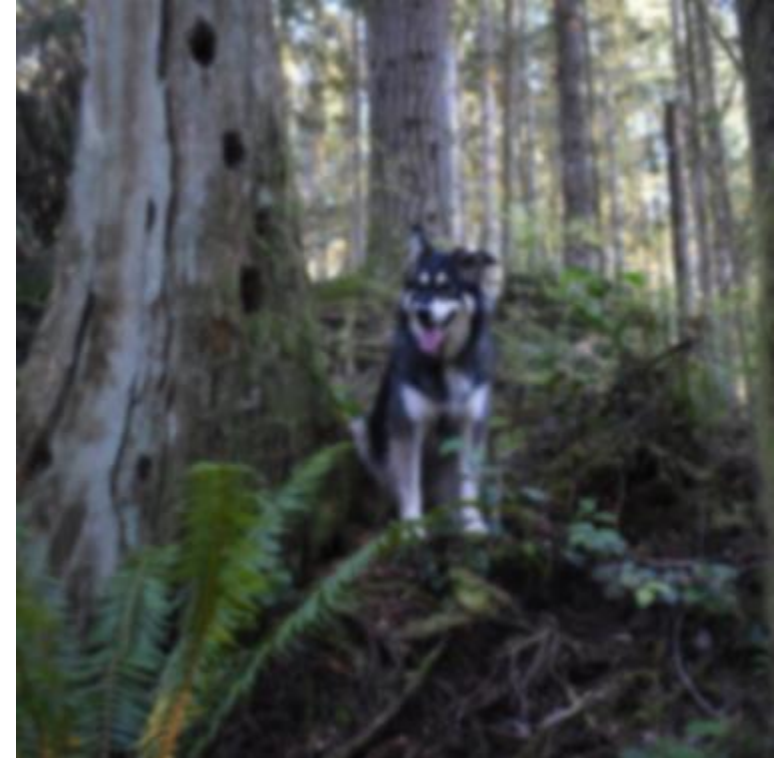
Better smoothing with Gaussians



*



=



Smoothing with box filter



Smoothing with Gaussian filter

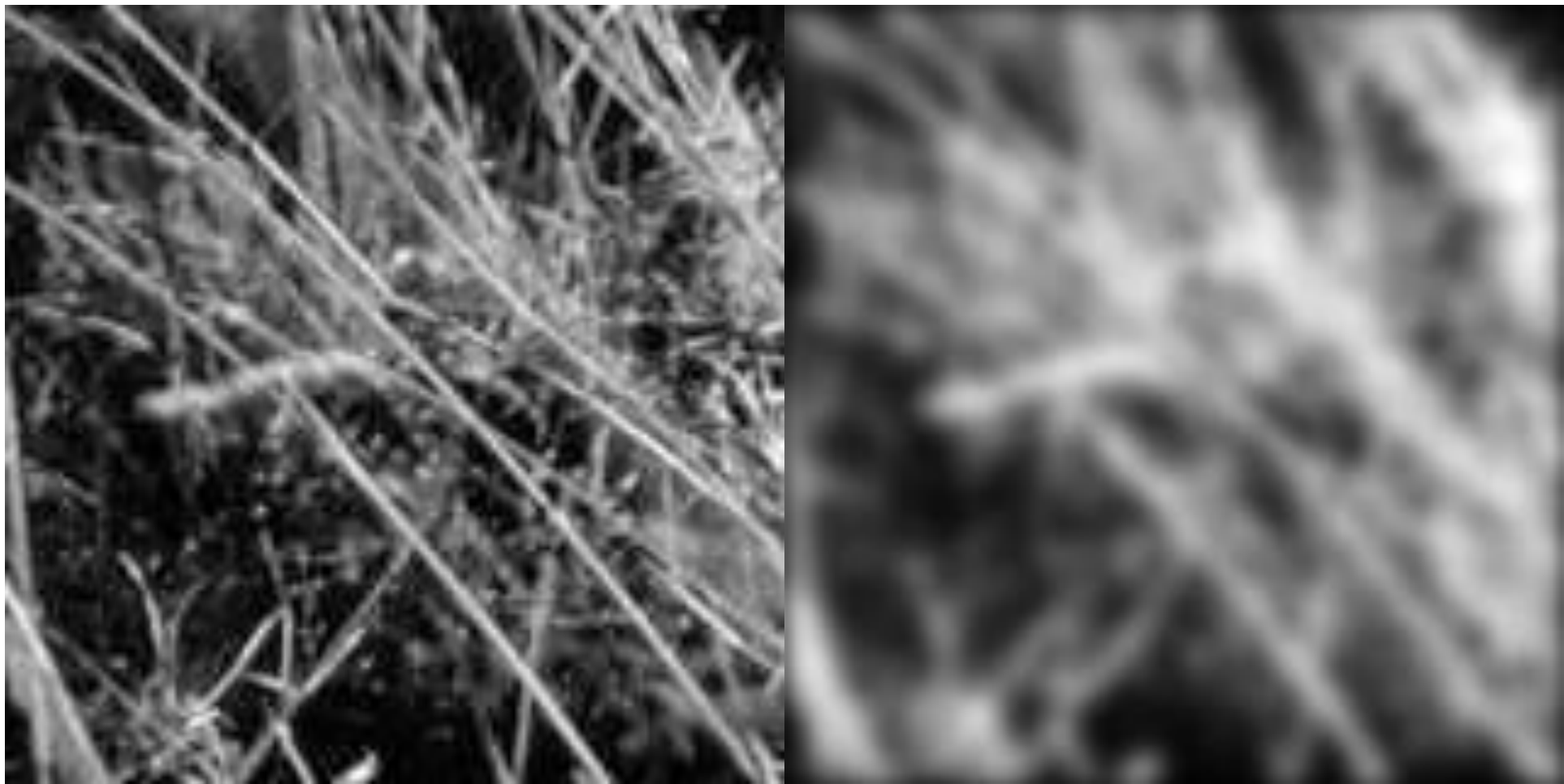


Image filtering – General definition

Compute function of
local neighborhood
at each position

$$\begin{array}{c}
 \text{h=output} \qquad \text{f=filter} \quad \text{I=image} \\
 \longrightarrow \quad h[m,n] = \sum_{k,l} f[k,l] I[m+k, n+l] \\
 \qquad \qquad \qquad \text{2d coords=k,l} \quad \text{2d coords=m,n} \\
 \begin{array}{ccc}
 \begin{bmatrix} \quad \end{bmatrix} & \begin{bmatrix} \quad \end{bmatrix} & \begin{bmatrix} \quad \end{bmatrix}
 \end{array}
 \end{array}$$

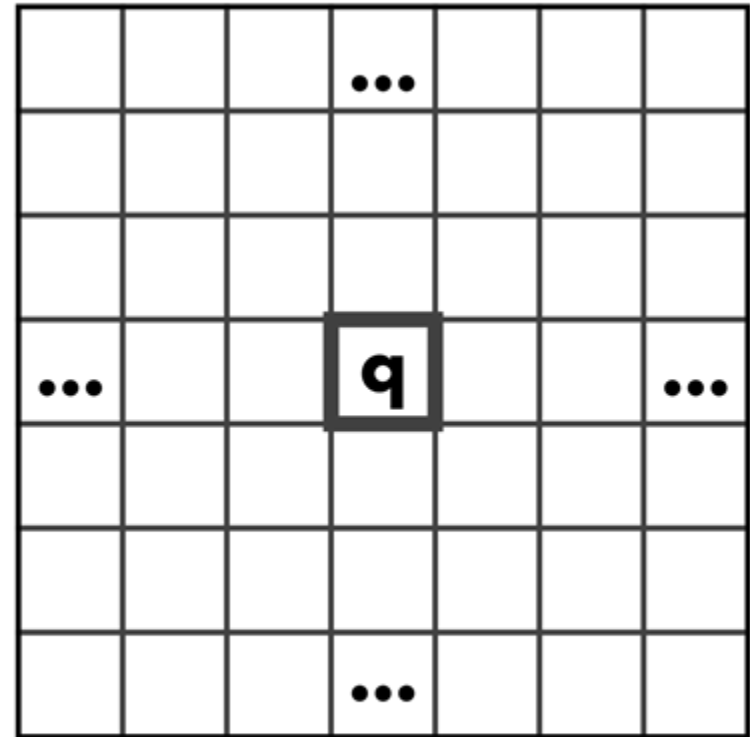
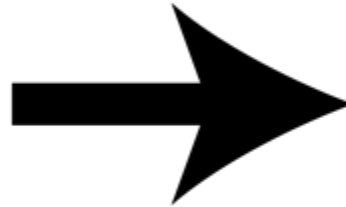
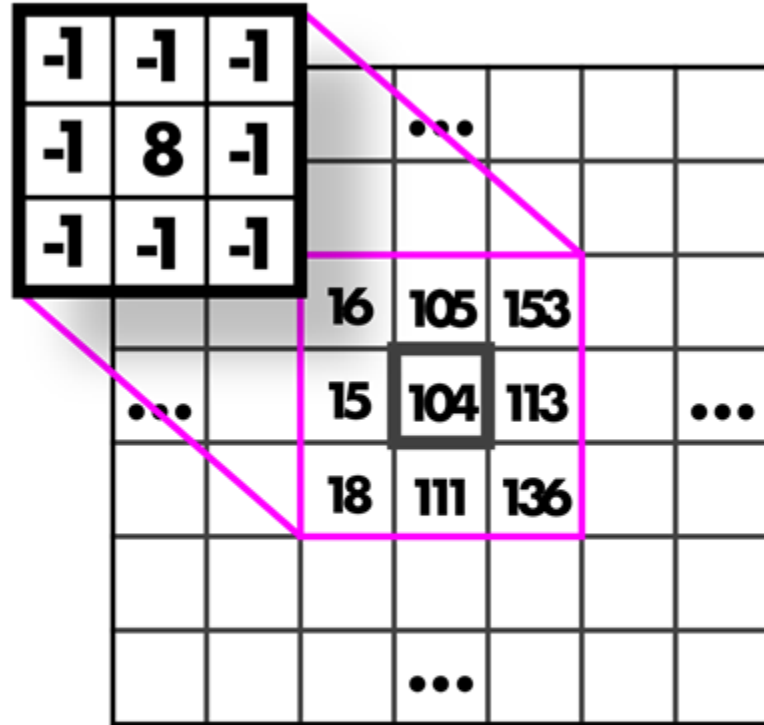
$$h = I * f$$

As mentioned before, this operation is often called ***convolution***

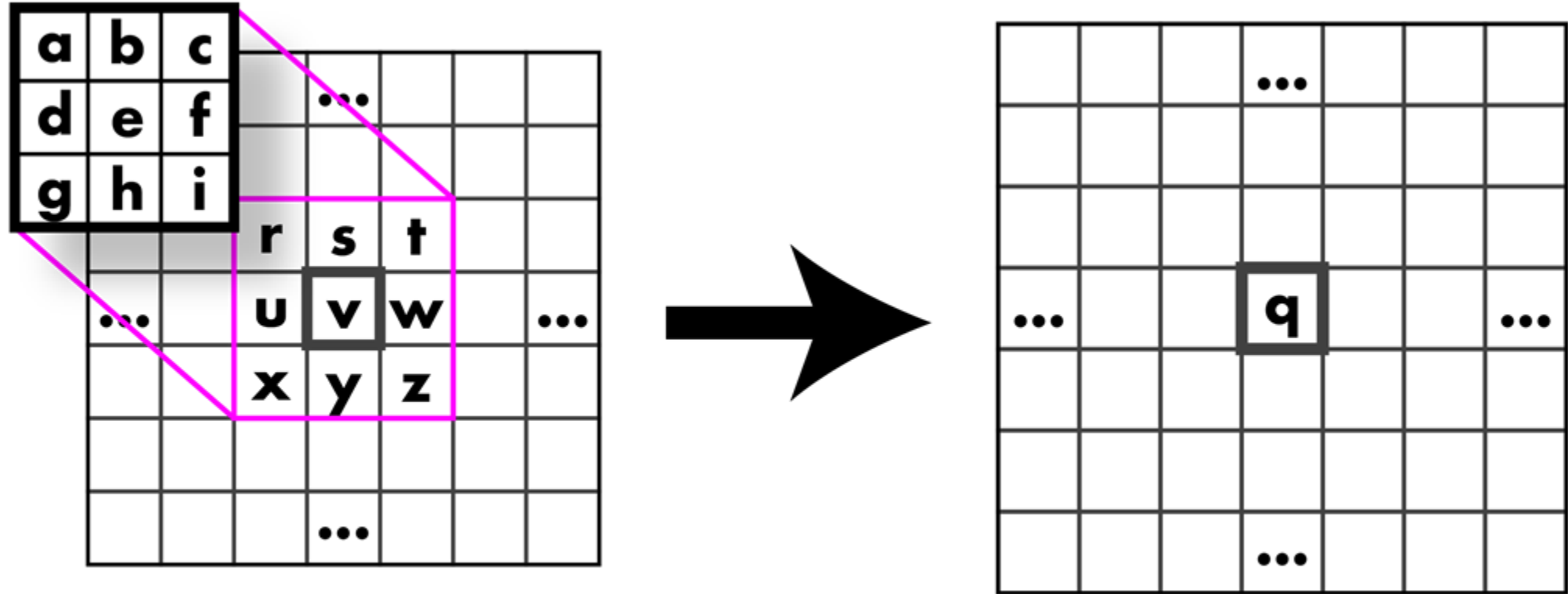
The filter is also called ***kernel***

Exercise

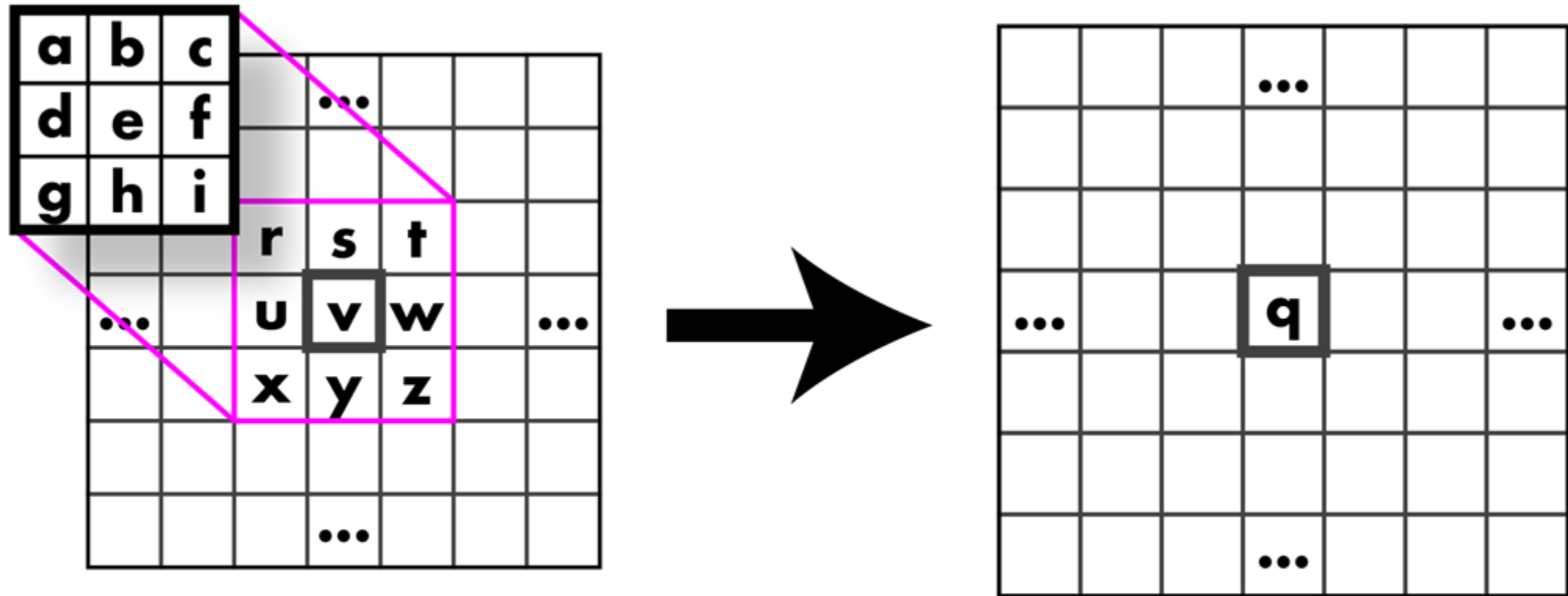
- $q = ?$



Convolution

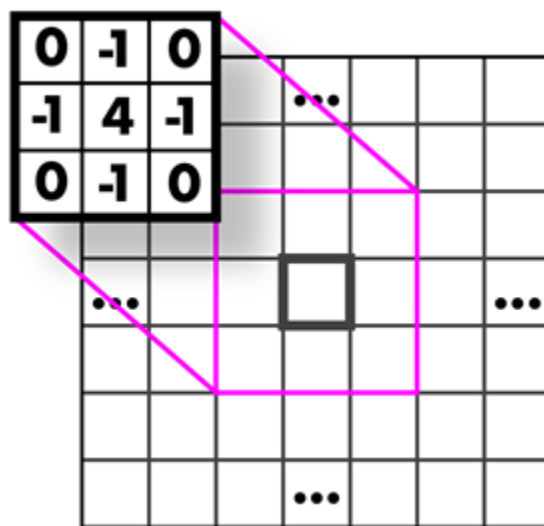


Convolution



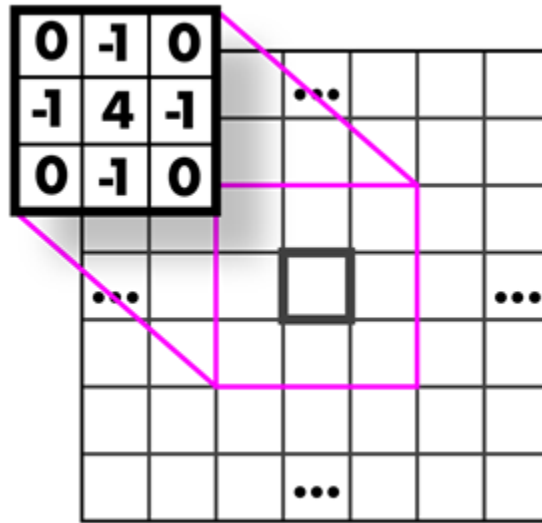
$$q = a \times r + b \times s + c \times t + d \times u + e \times v + f \times w + g \times x + h \times y + i \times z$$

What is this kernel?

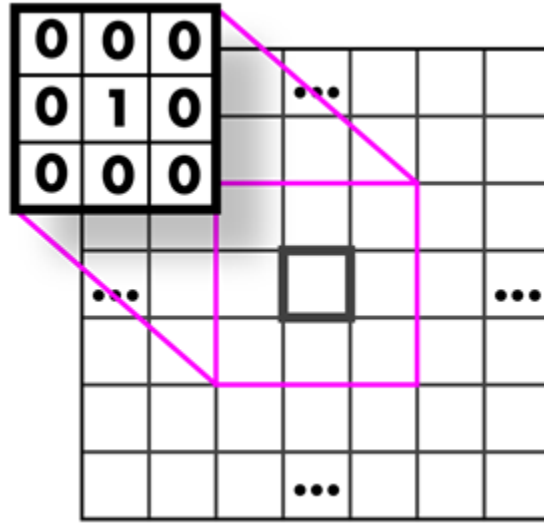


unlike the box filter,
this filter sums to 0

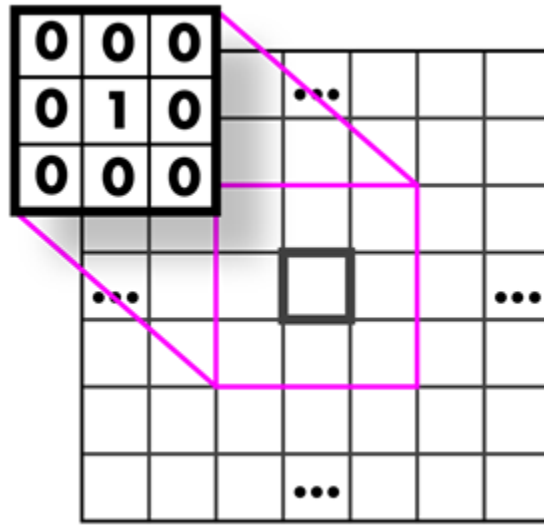
Highpass kernel $\rightarrow$ finds edges



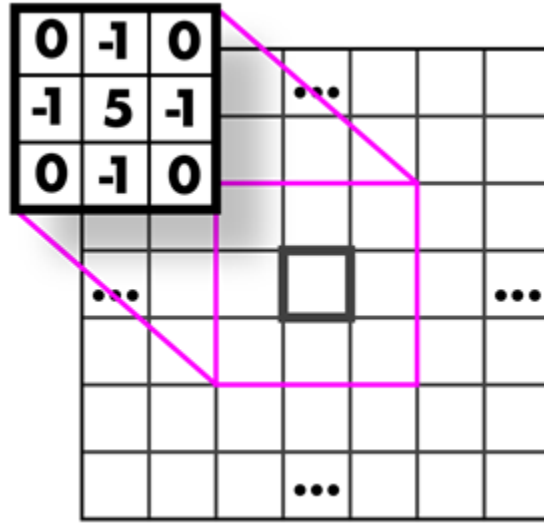
What is this kernel?



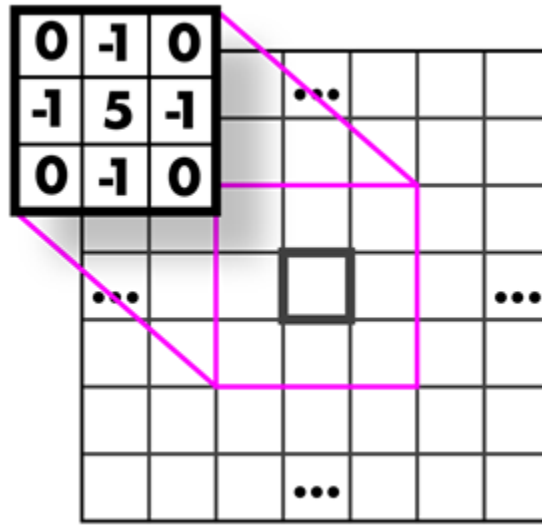
Identity kernel $\rightarrow$ does nothing



What is this kernel?

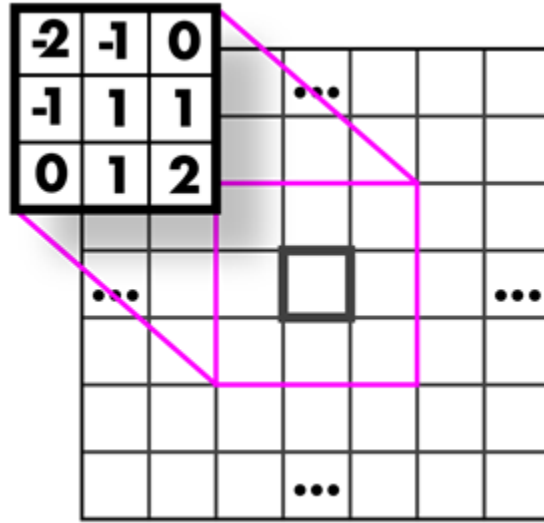


Sharpen Kernel $\rightarrow$ sharpens image

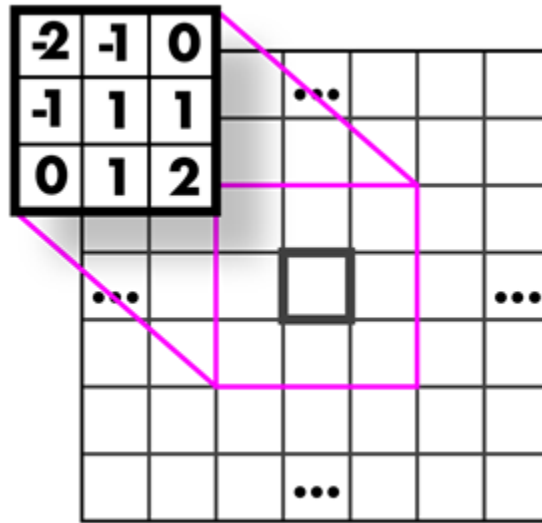


Note: sharpen = highpass + identity!

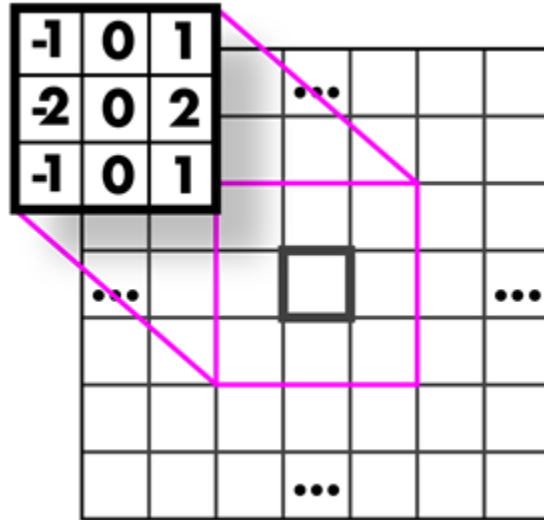
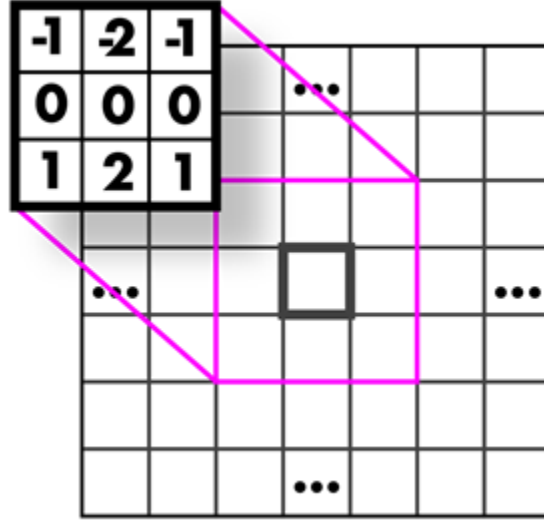
What is this kernel?



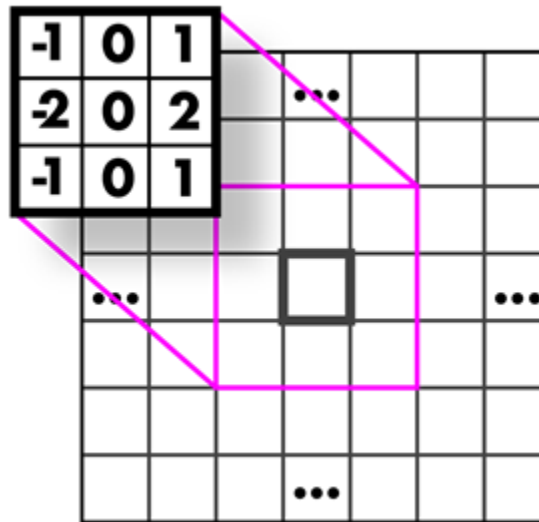
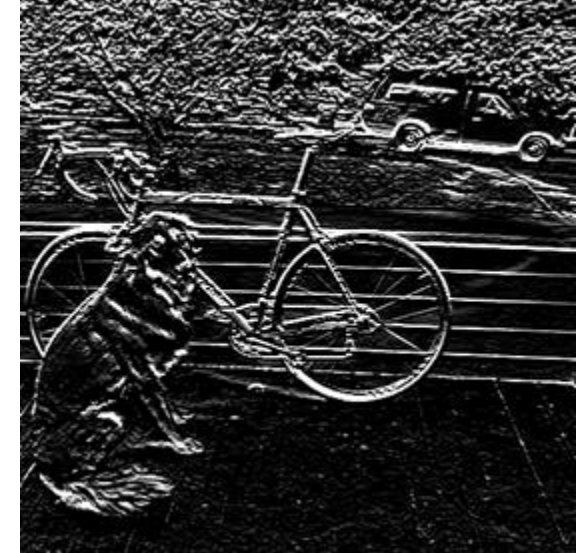
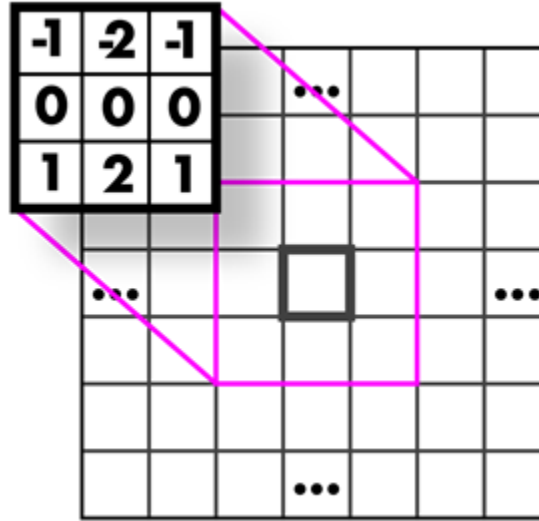
Emboss Kernel → stylizes image



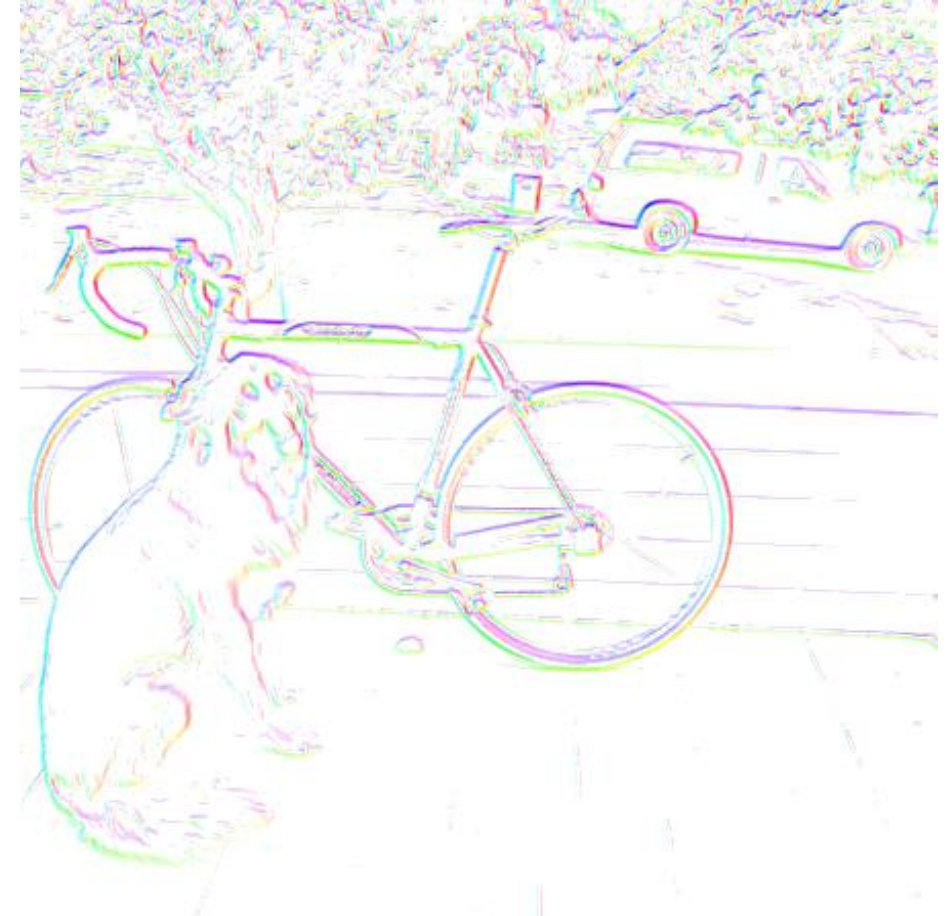
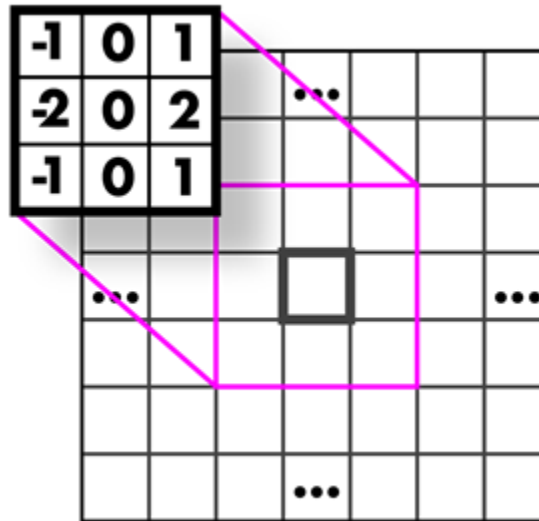
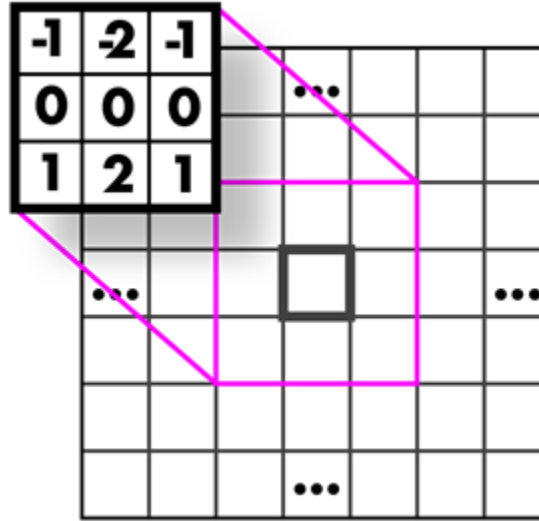
What are these kernels?



Sobel Kernels $\rightarrow$ edges and...



Sobel Kernels $\rightarrow$ edges and gradients

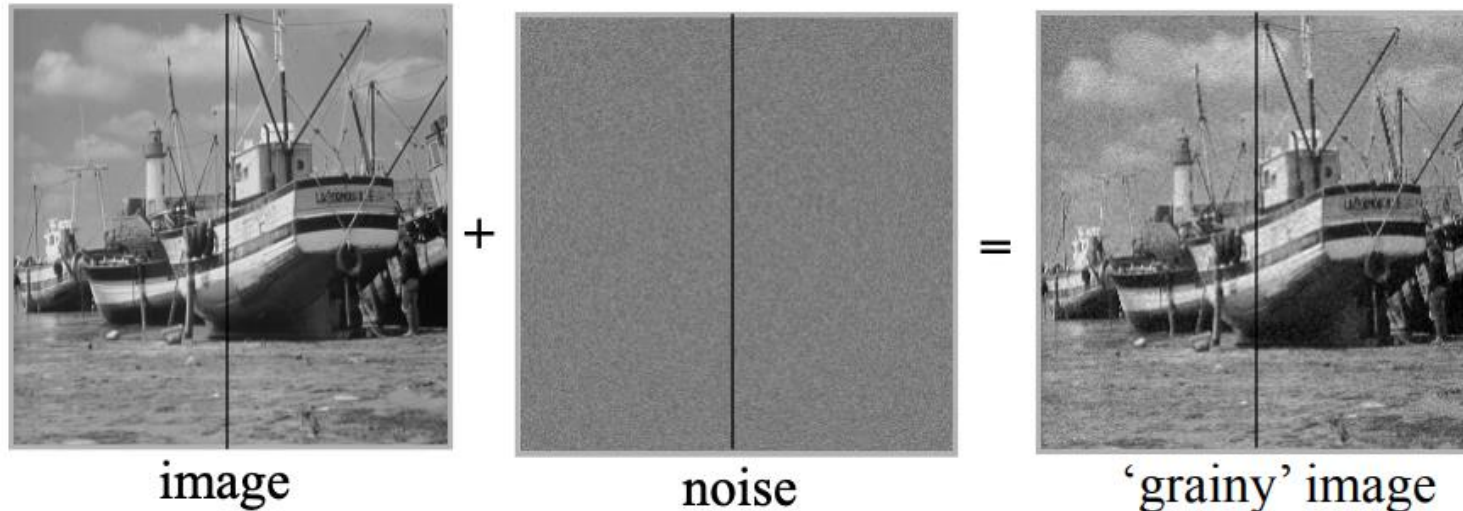


Noise reduction

an application of filtering

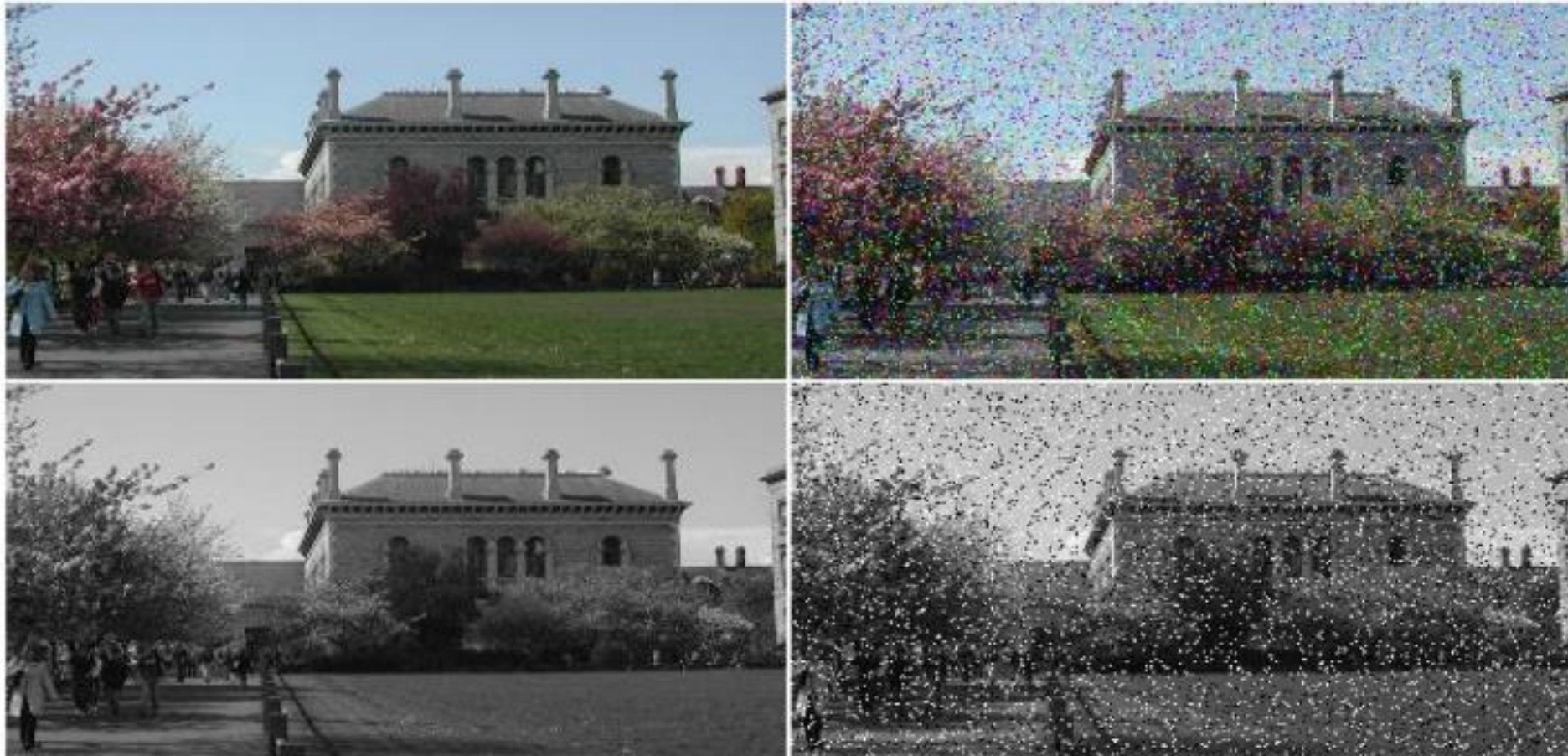
What is noise?

- Image corruption
- Noise fluctuations are rapid $\rightarrow$ high frequency
- Sources of noise
 - CCD chip $\rightarrow$ electronic signal fluctuations in detector
 - Other electronics
 - Transmission (analog)



Noise – salt & pepper

- Impulse noise → noise is maximum or minimum values



source: A Practical Introduction to Computer Vision with OpenCV , by Kenneth Dawson-Howe

Noise – Gaussian

- Good approximation to real noise → Gaussian distribution



source: A Practical Introduction to Computer Vision with OpenCV , by Kenneth Dawson-Howe

Noisy – Salt and Pepper



Mean – 3 x 3 filter



Mean – 11 x 11 filter



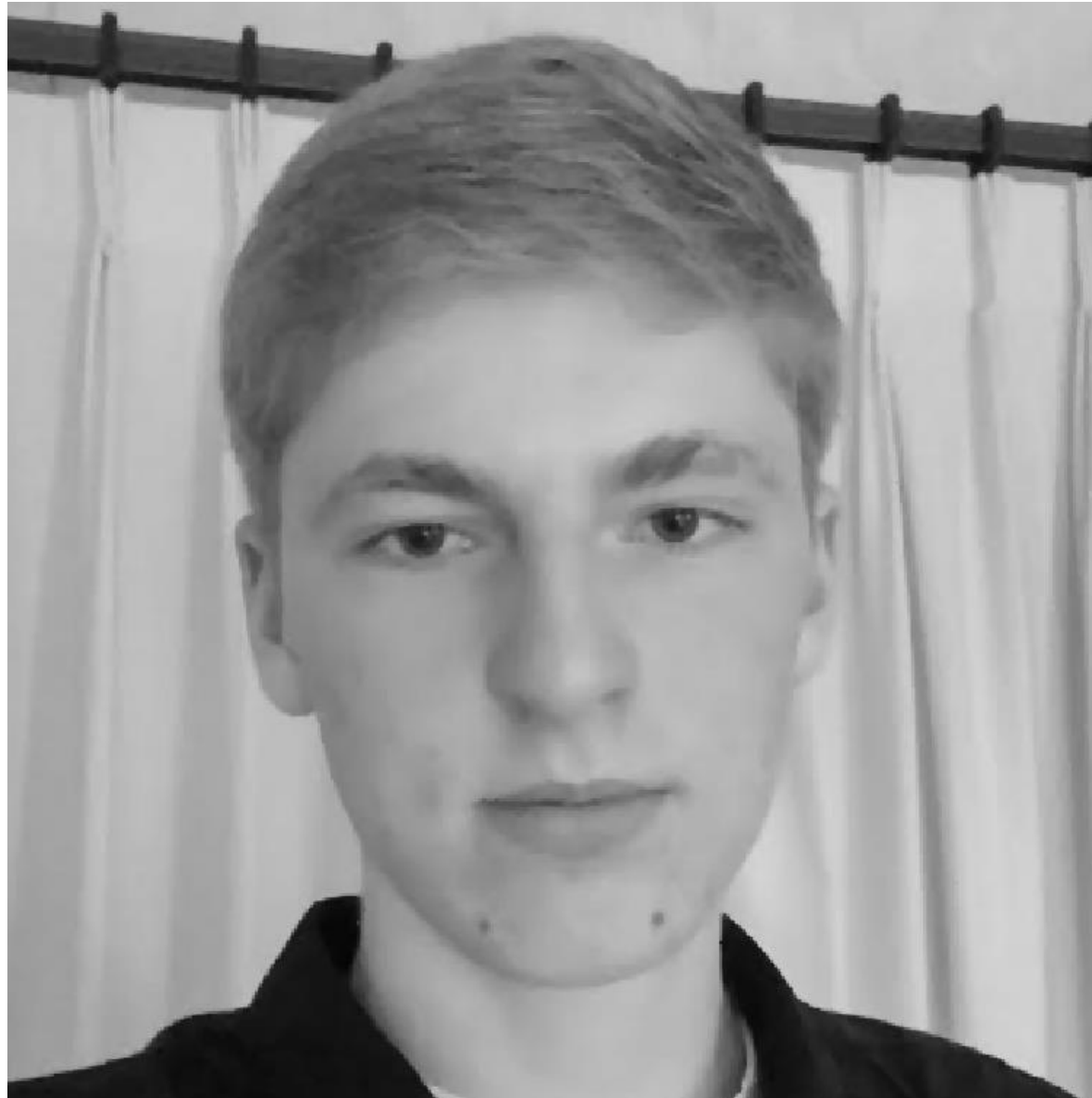
Median filters

- Operates over a window by selecting the **median intensity** in the window.
- ***Rank filter*** as based on ordering of gray levels
 - e.g., min, max, range filters
- Not a convolution-based filter

Noisy – Salt and Pepper



Median – 3 x 3



James Tompkin

Noise reduction

- Removing (is it completely possible?) or reducing noise
- Linear smoothing transformations
 - frame averaging → acquire several frames of static scene
 - local image filtering (average, Gaussian)
- Non-linear transformations
 - median filter
 - anisotropic diffusion filter, bilateral filter, etc (*)
- Frequency domain filters (*)

(*) Outside of the scope of this course, but if you want to learn more about noise reduction, check the extra slides in Moodle